Analysis group meeting

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07/21/25

MISSISSIPPI STATE UNIVERSITY

RG-D Pass1

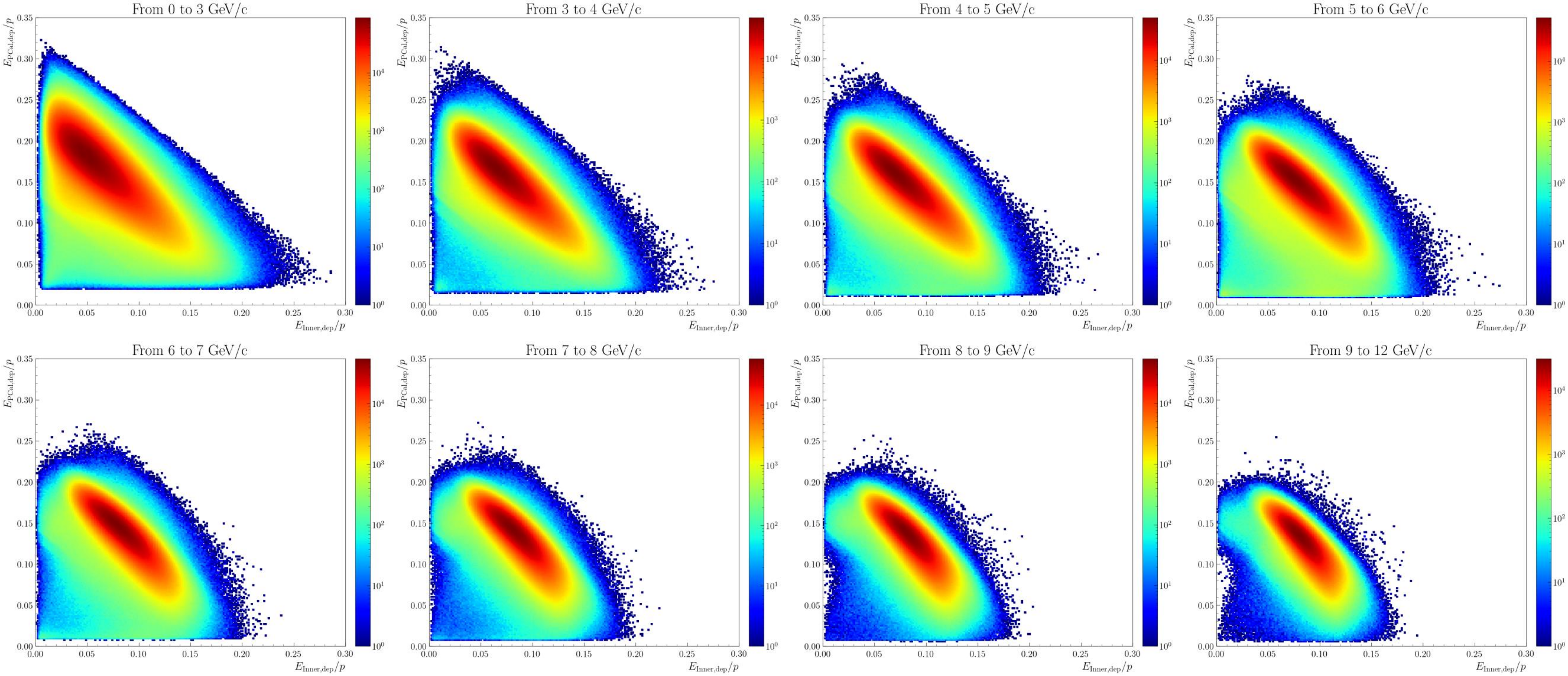
workflow_name	jobs	success
rgd-dmra-pass1_CuSn_1-18561x33	14687	5469 (37.24%)
rgd-dmra-pass1_CxC_1-18445x13	6873	6849 (99.65%)
rgd-dmra-pass1_LD2_1-18431x16	7702	7672 (99.61%)
rgd-dmra-pass1_empty_nominal-18399x2	113	70 (61.95%)
rgd-dmra-pass1_empty_zero_field-18316	239	154 (64.44%)
rgd-dmra-pass1_LD2_2-18305x144	40076	0 (0.00%)
rgd-dmra-pass1_CxC_2-18339x116	36705	0 (0.00%)
rgd-dmra-pass1_CuSn_2-18347x294	70791	0 (0.00%)

For all the running workflows (LD2_1, CxC_1, CuSn_1, empty_zero, empty_nominal):
Total: 29614, Done: 20214, Running: 5144, Pending: 4256
(Done/Total): 68.26%, (Running/Total): 17.37%, (Pending/Total): 14.37%

For all the workflows:
Total full: 177186.00, Done/Total full: 11.41%

RG-D: Fiducial cuts

- The sampling fractions of the inner calorimeter and the PCAL: Outbending LD2



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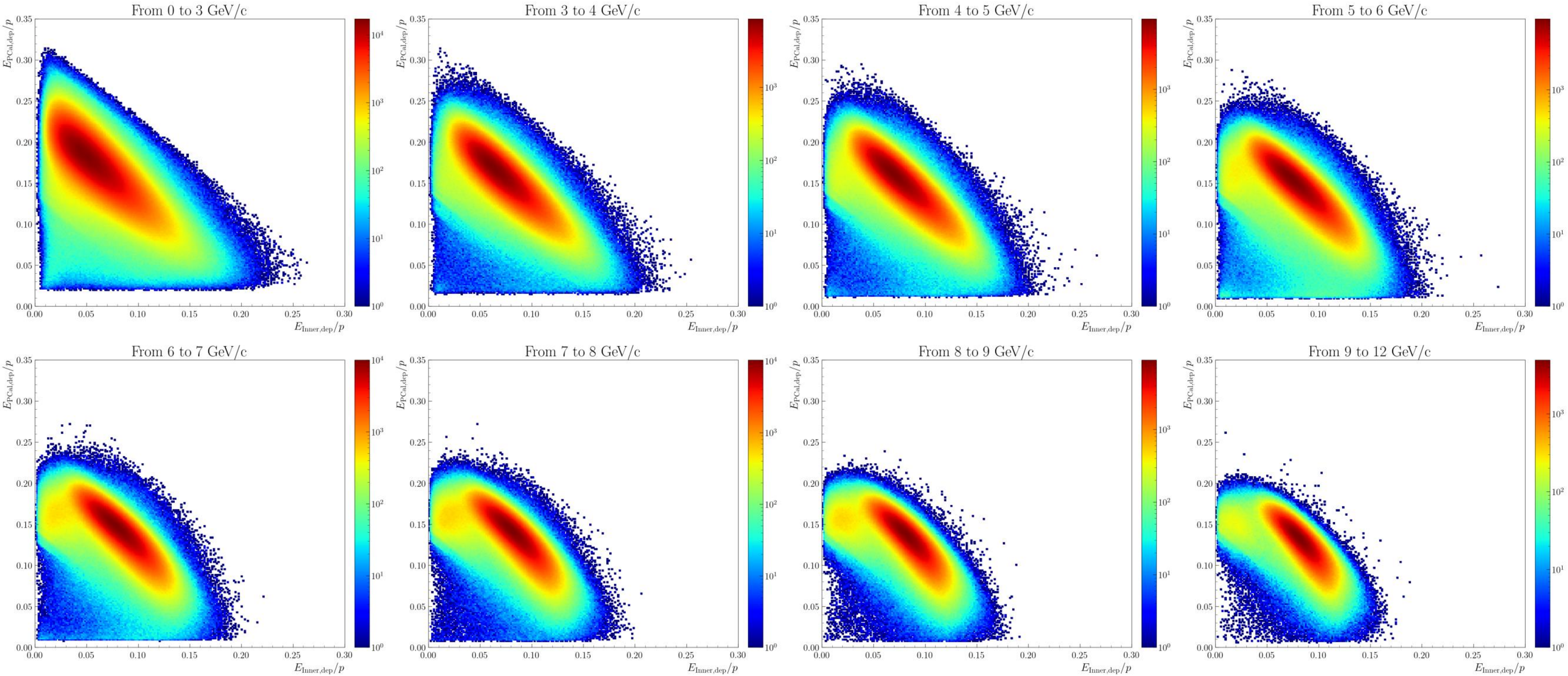


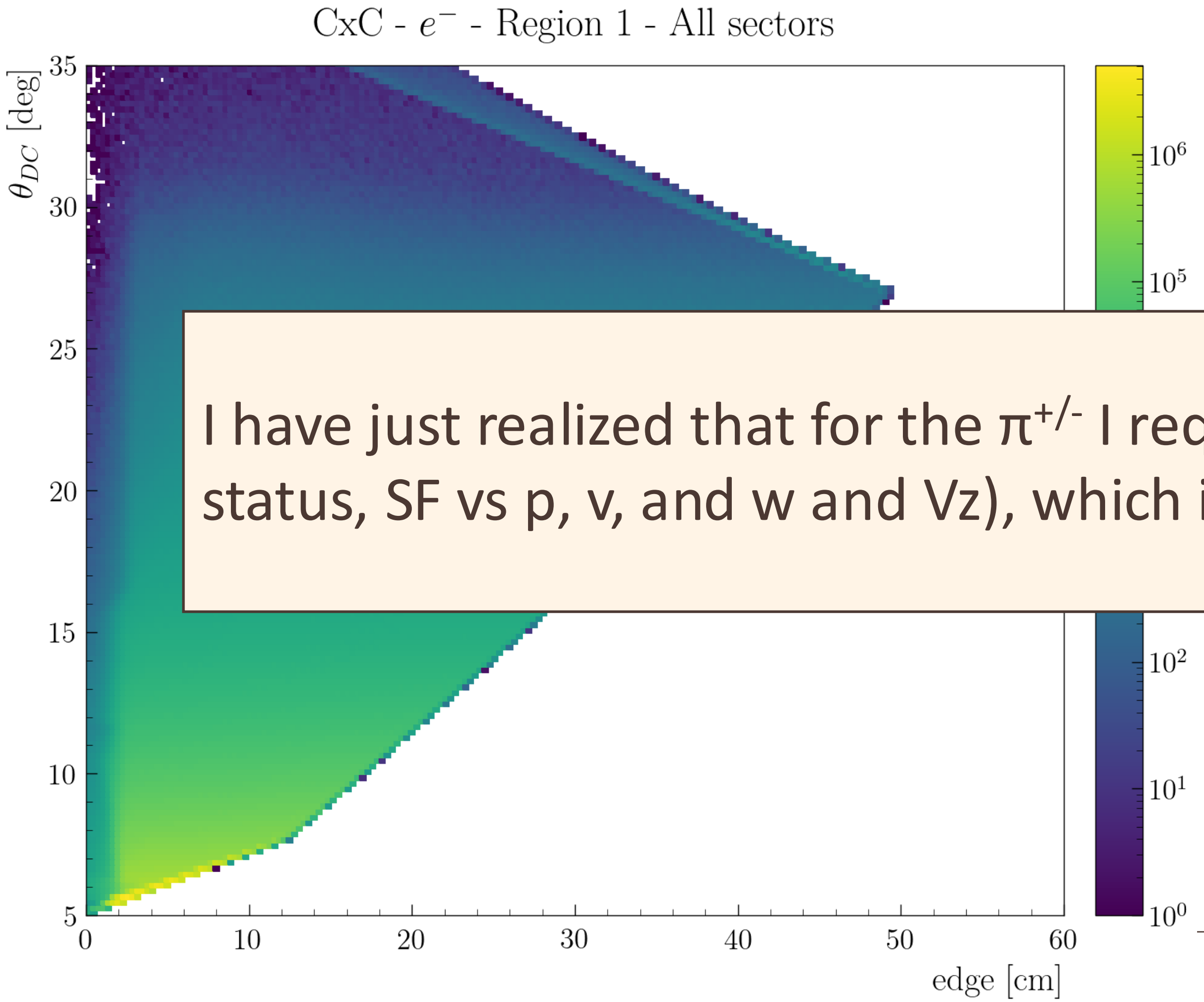
RG-D: Fiducial cuts

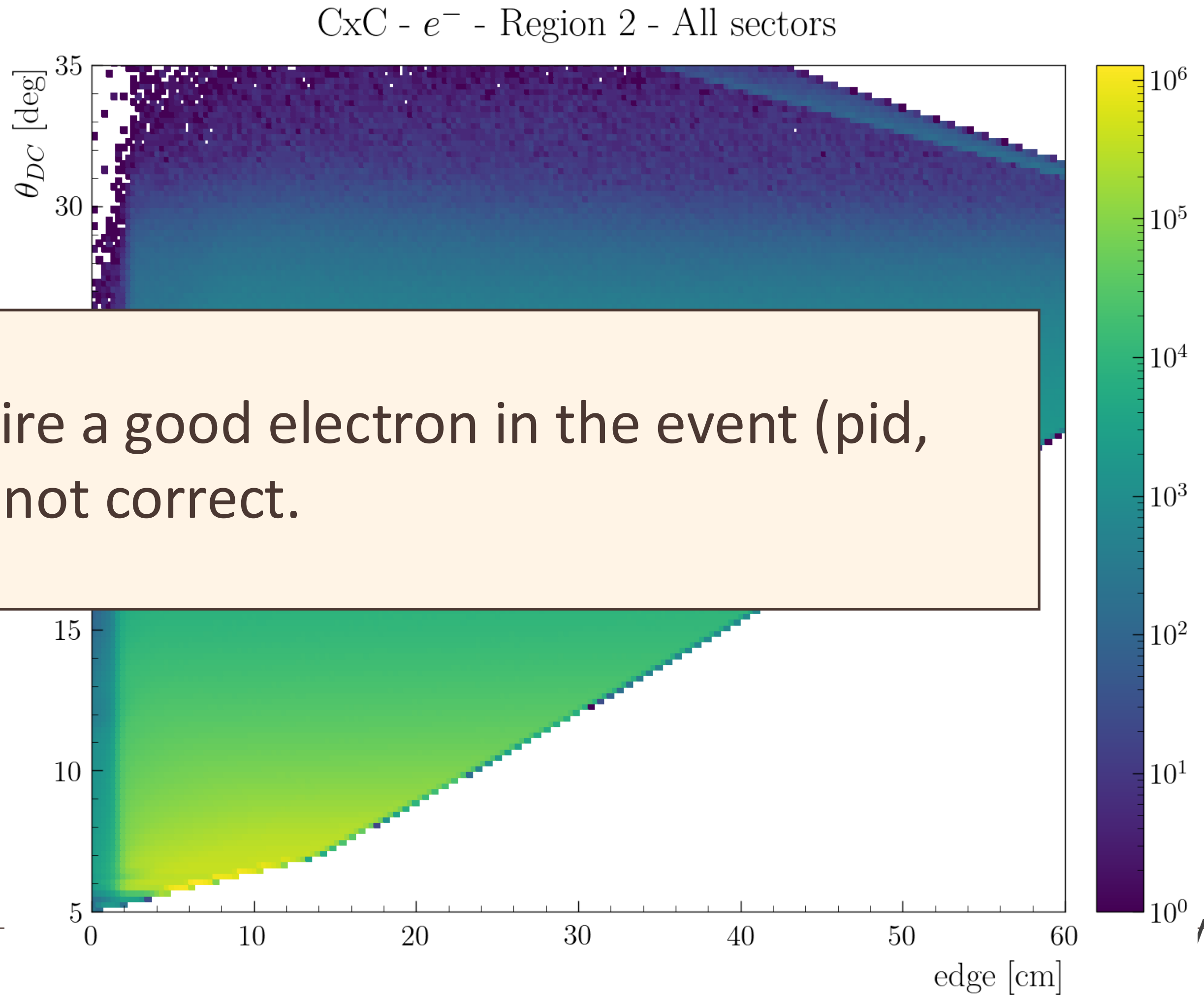
- Clear contamination in the inbending case, maybe in the outbending:
- Develop a cut for both:
  - Slice the histograms for each momentum bin, fit the distribution with a Gaussian, and extract the mean and sigma.
  - Fit the mean +/- sigma with this function:

$$\frac{E_{PCAL}}{p}/p \;>\; \left(p1\_secX[p\_par] \;-\; p0\_secX[p\_par] \;\cdot\; \frac{E_{ECin}}{p}\right),$$

RG-D: DC fiducial cut

- I have discussed with Raffaella about the DC fiducial cut:
  - Her advice is to develop particle-independent cuts
  - For outbending and for regions 1 and 2, we can use electrons/π^- since they fill the histograms





I have just realized that for the $\pi^{+/-}$ I require a good electron in the event (pid, status, SF vs p, v, and w and Vz), which is not correct.

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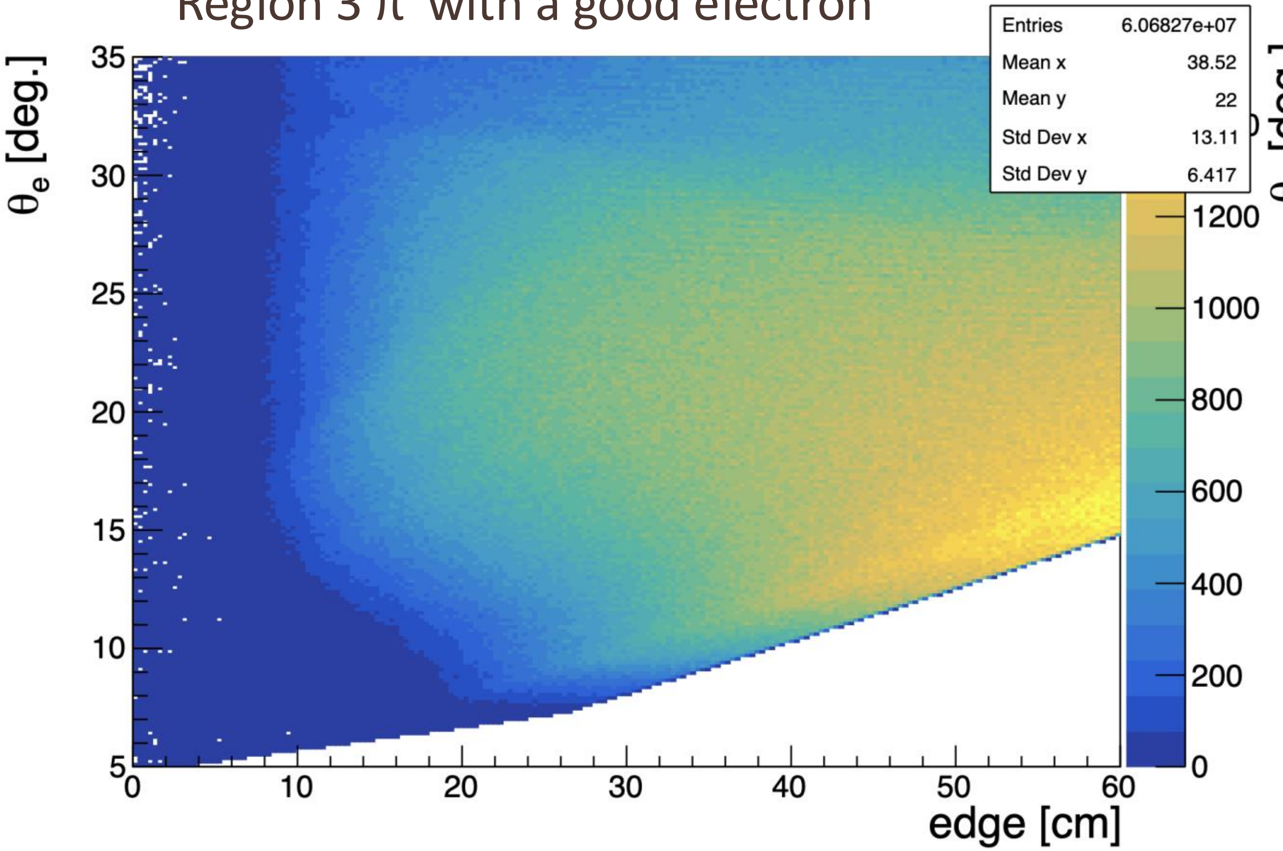
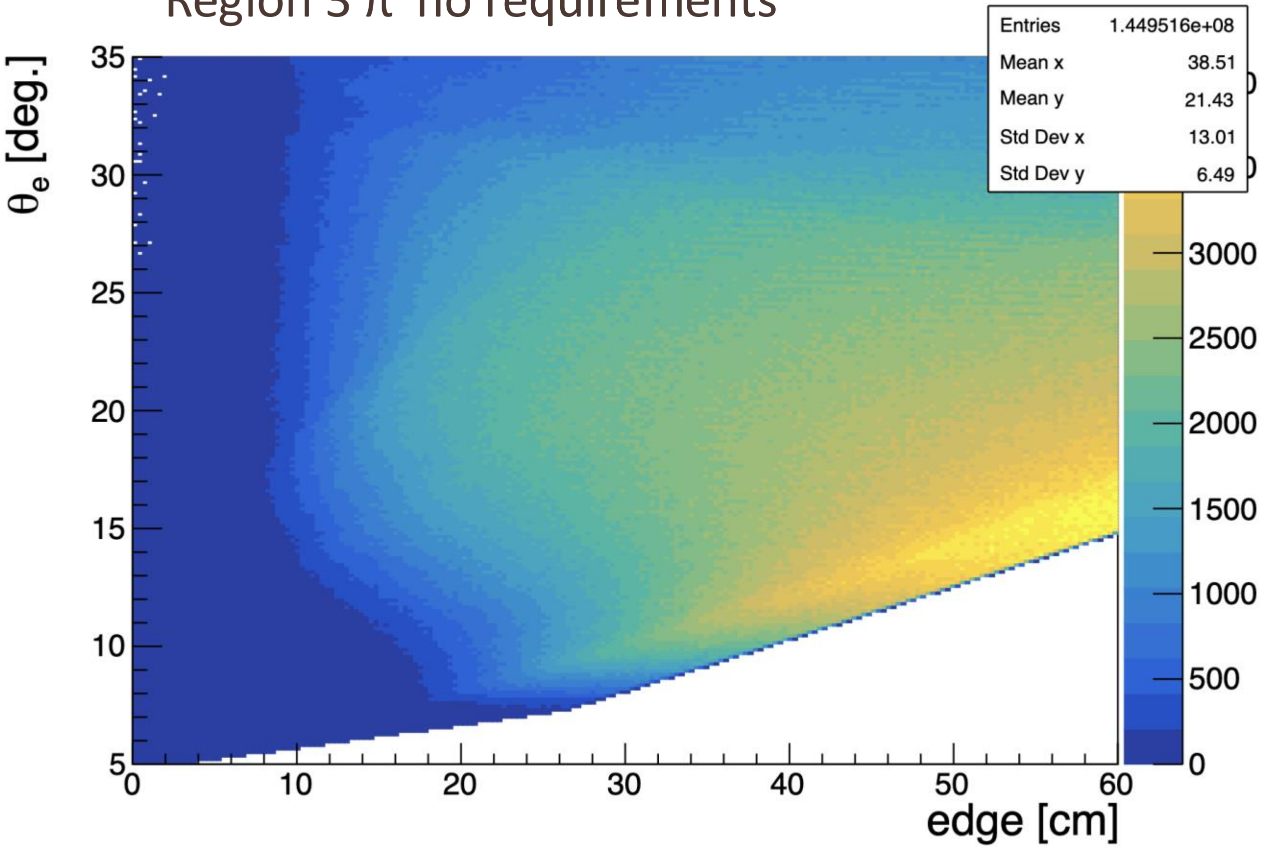
Region 3 π^- with a good electron



Region 3 π^- no requirements



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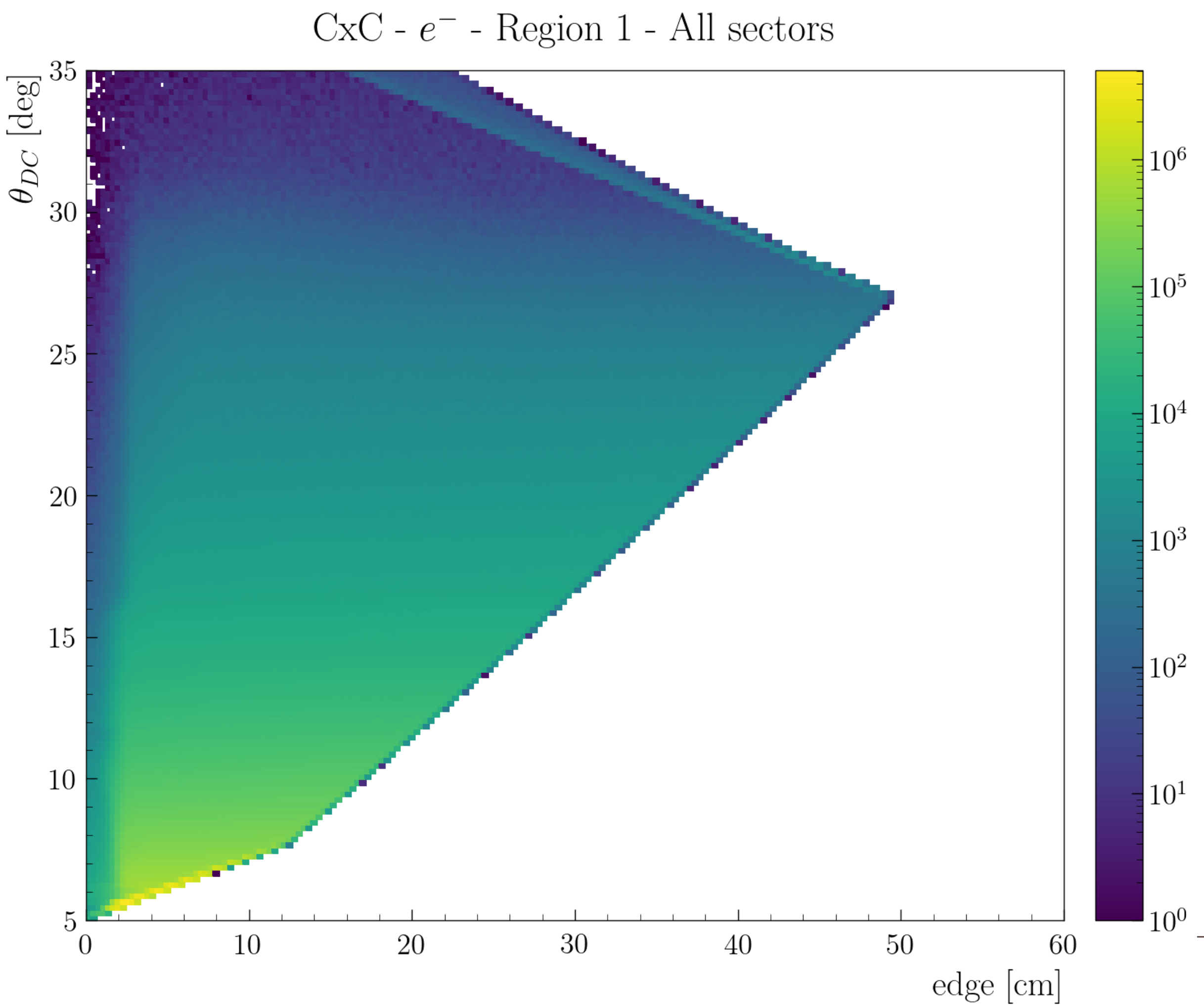

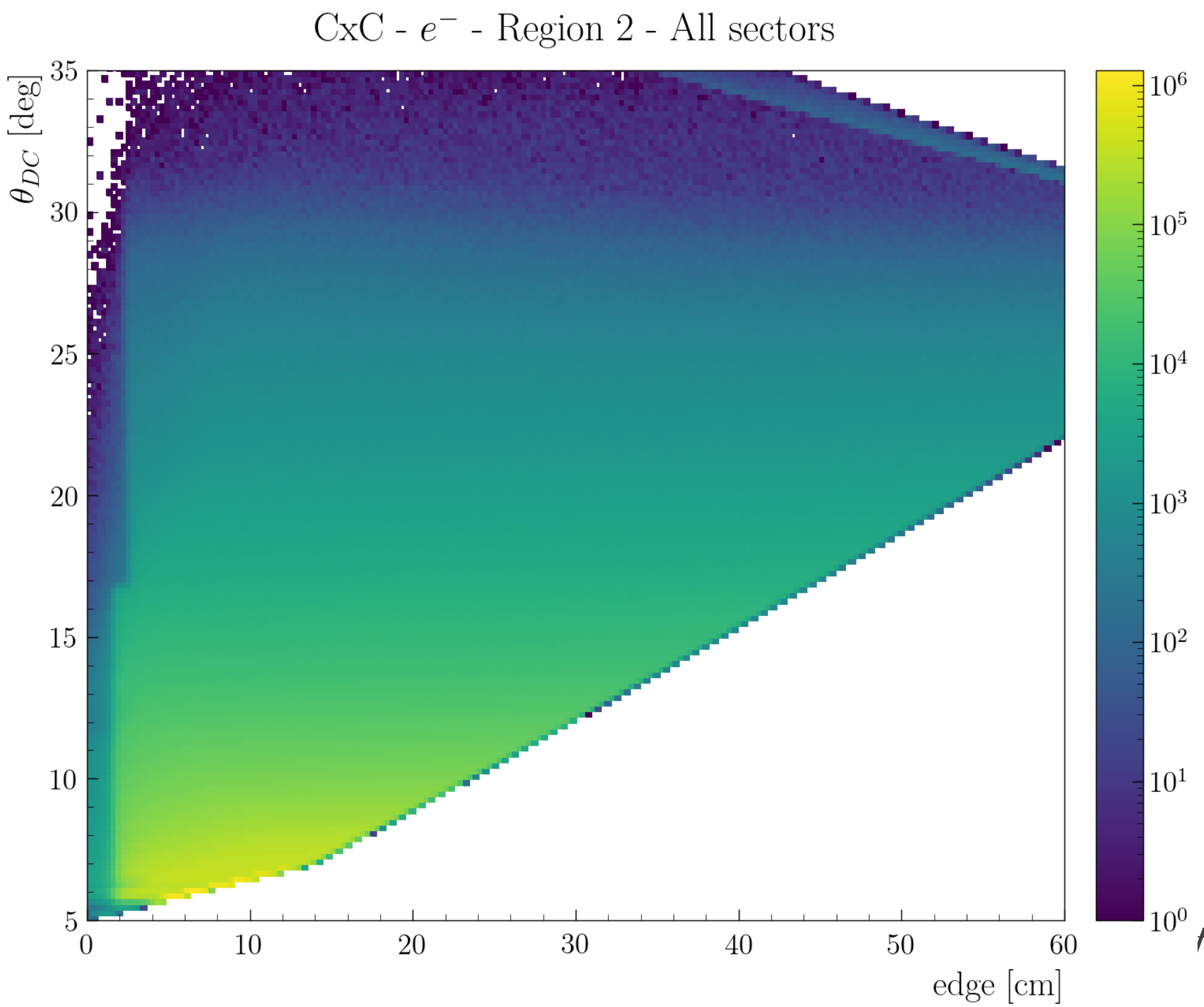

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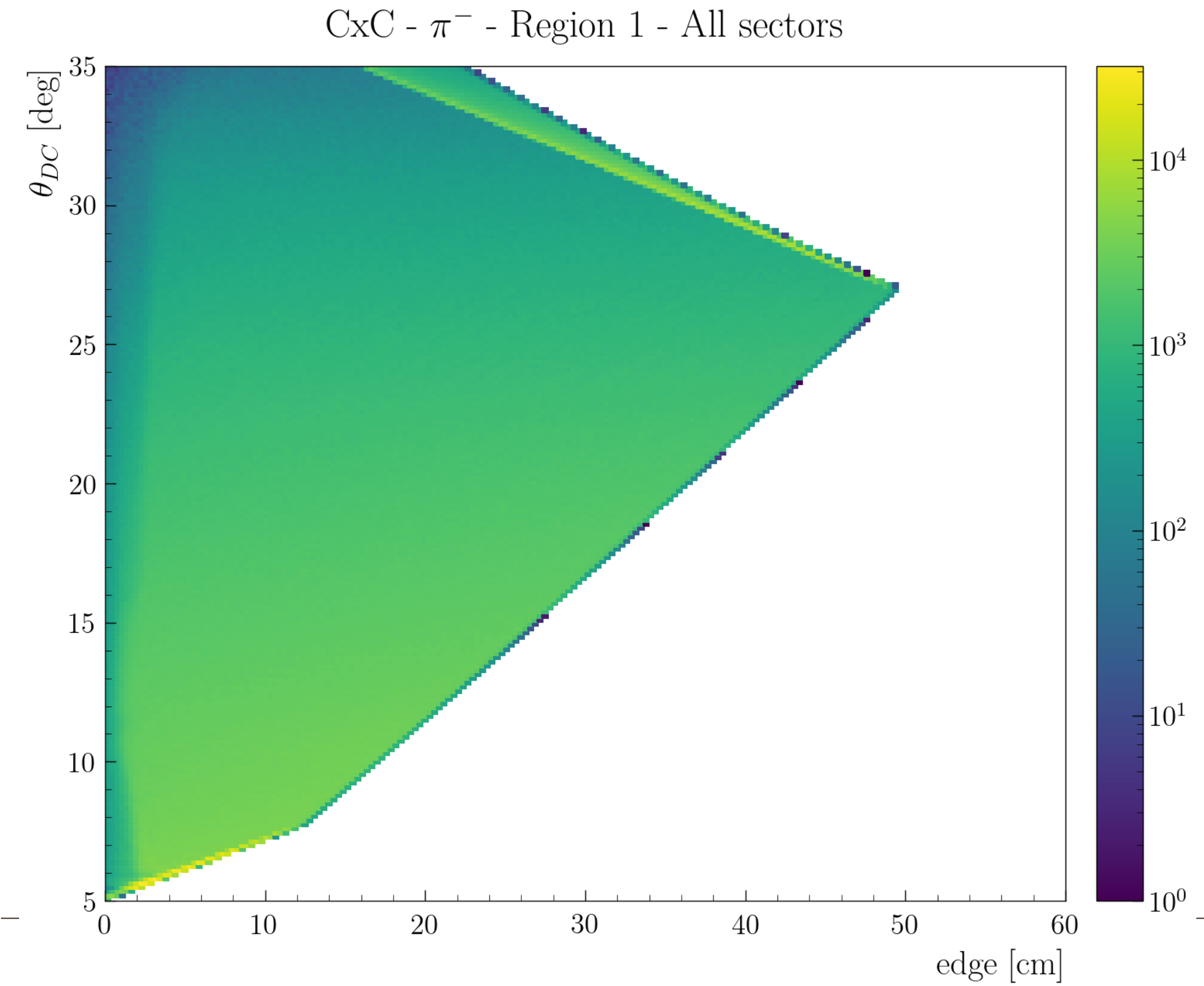

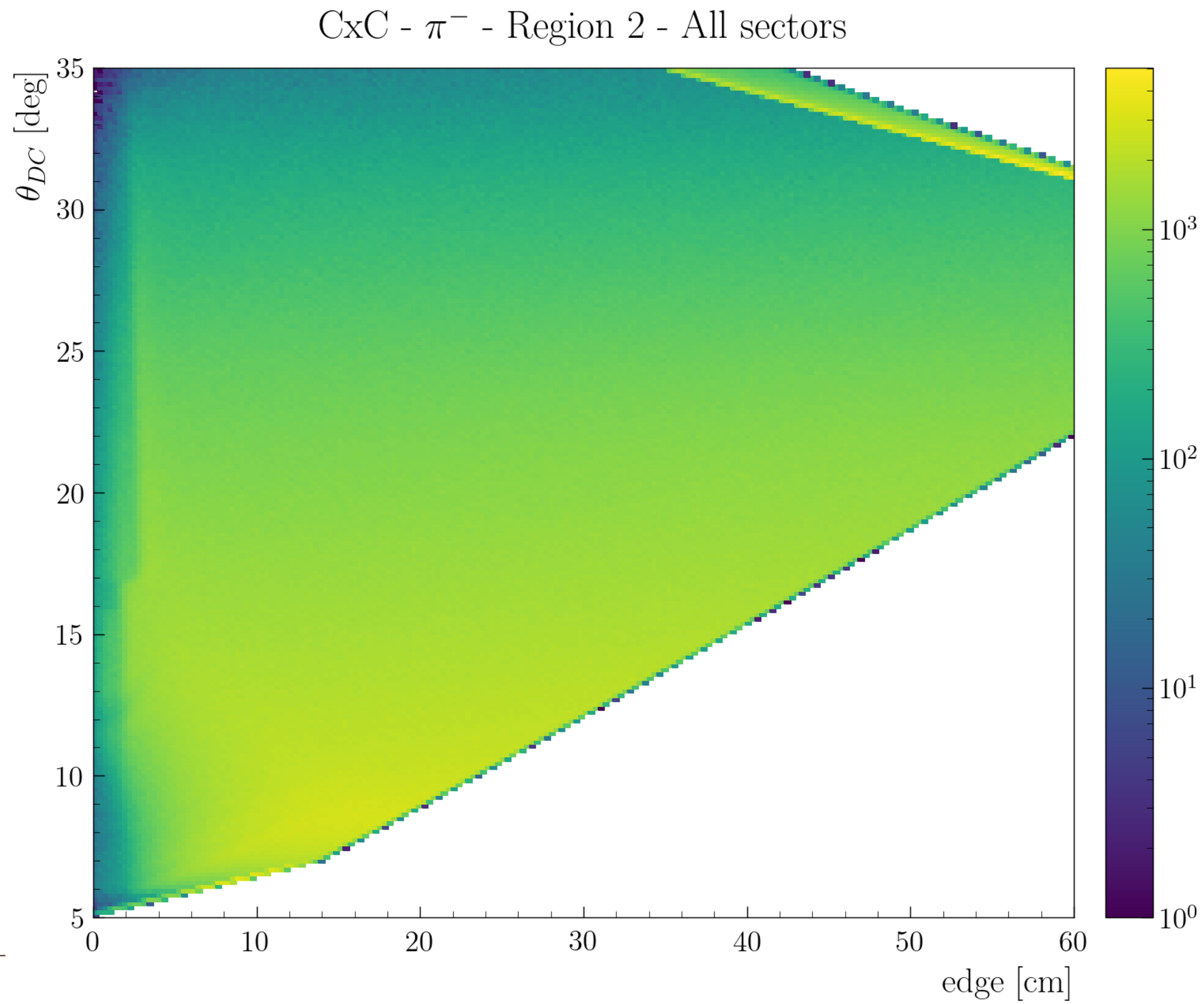

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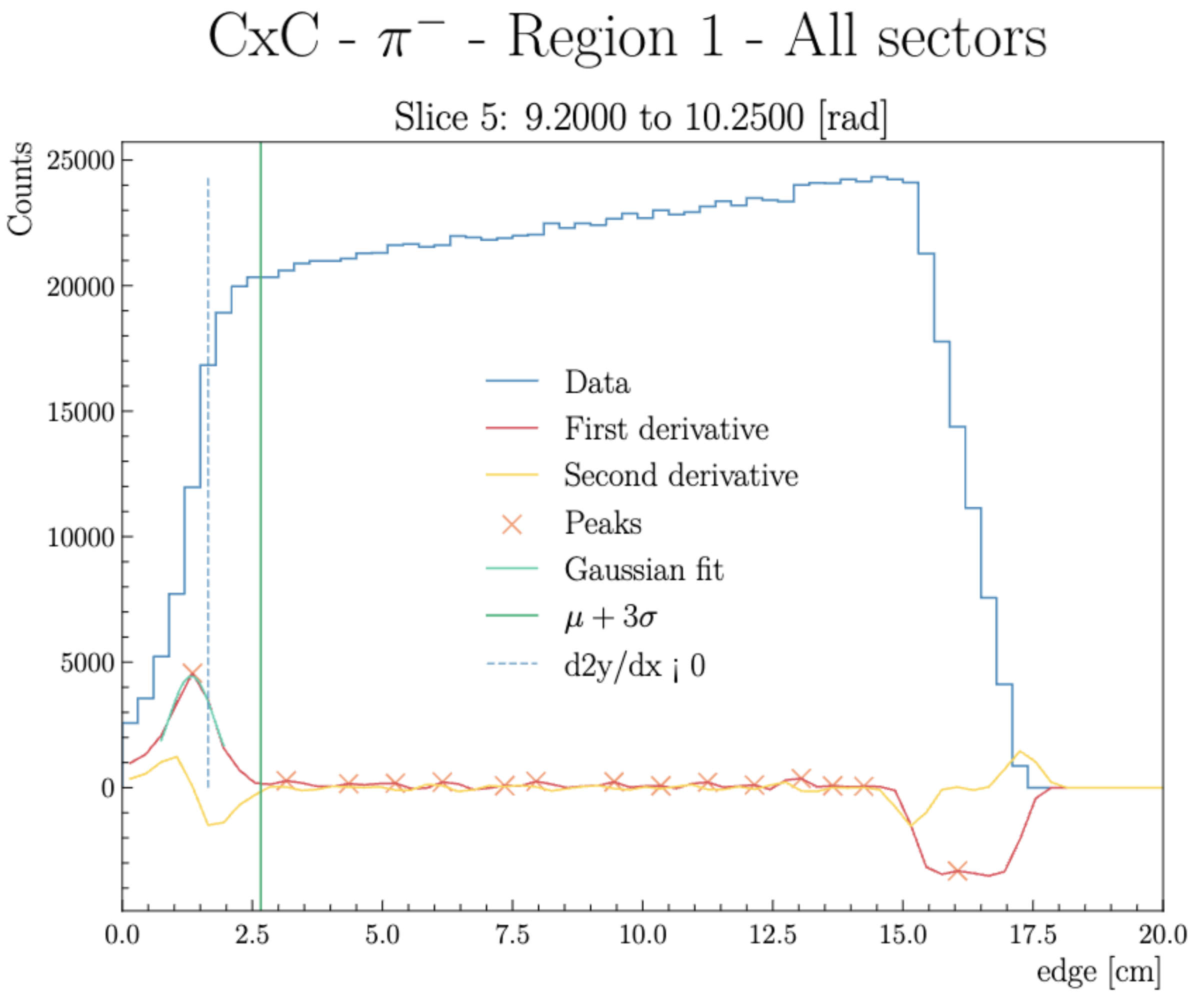

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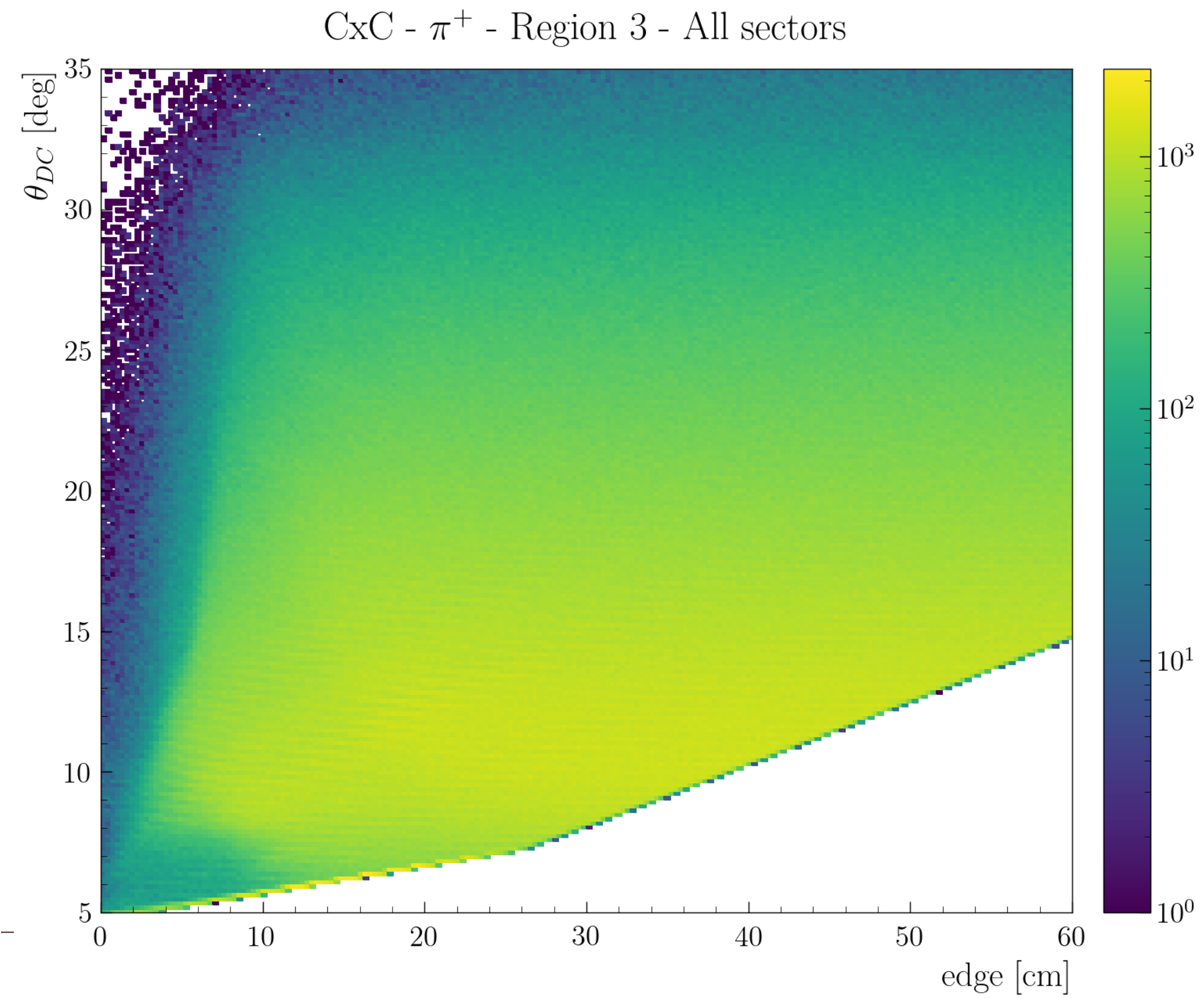

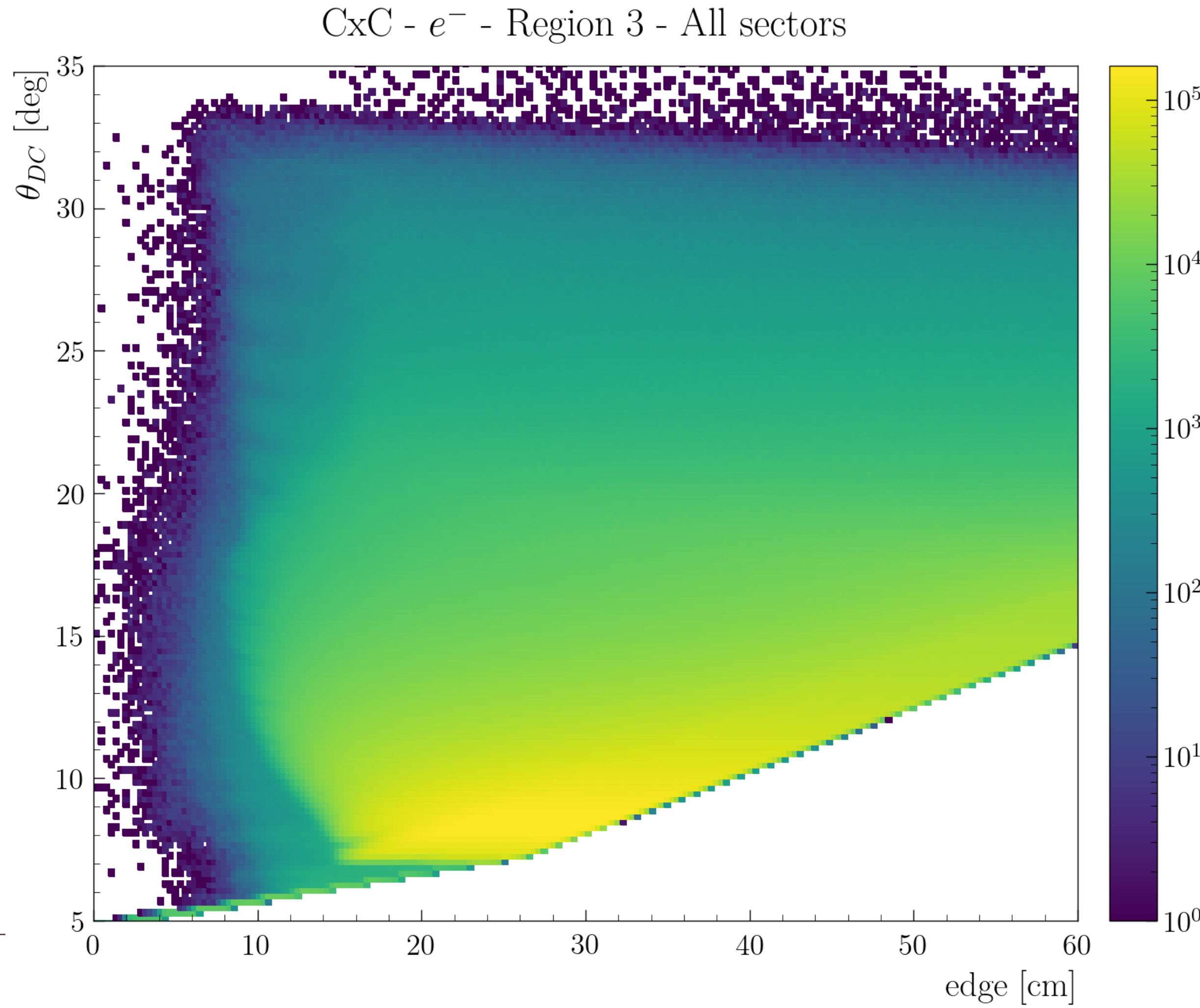

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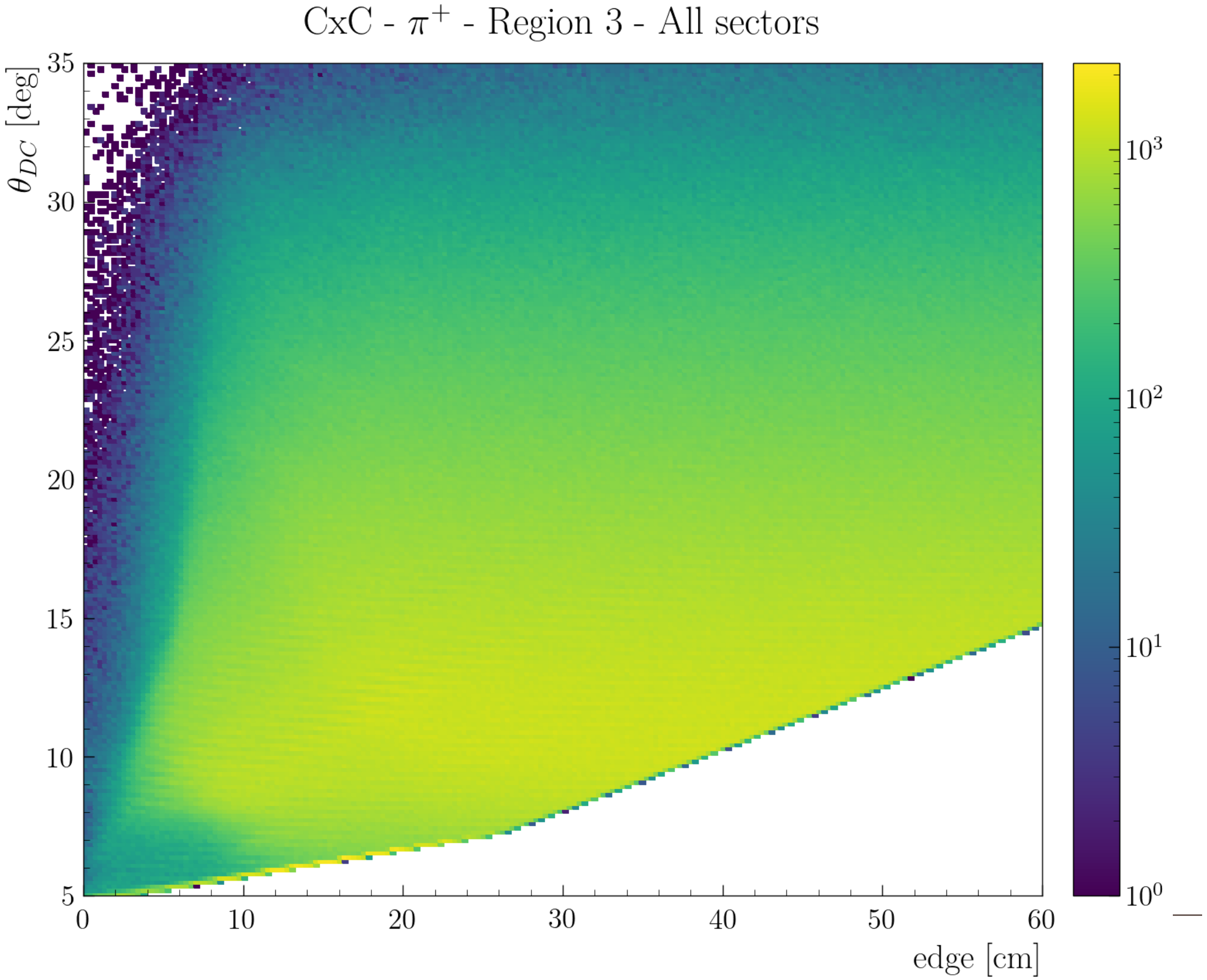

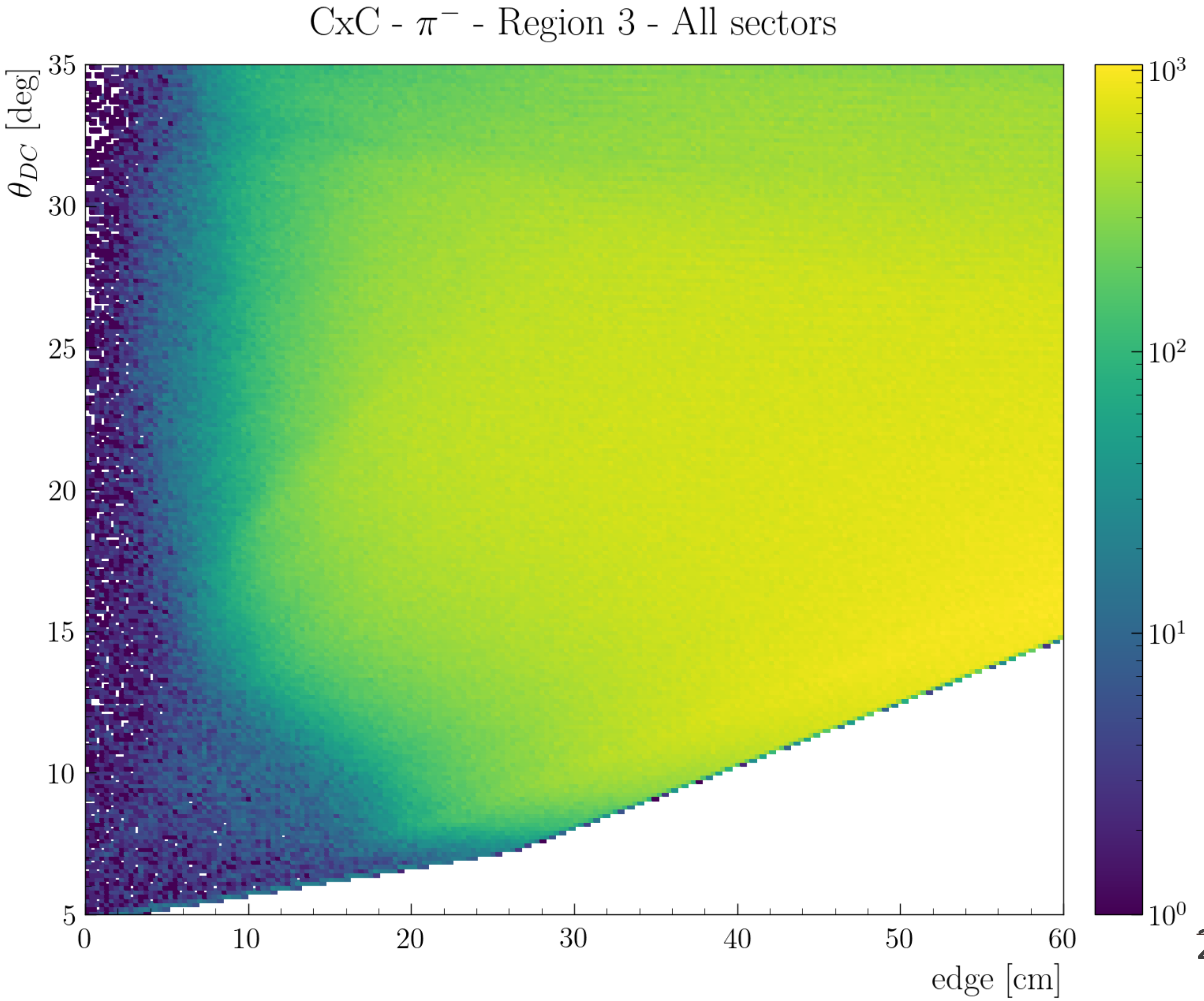

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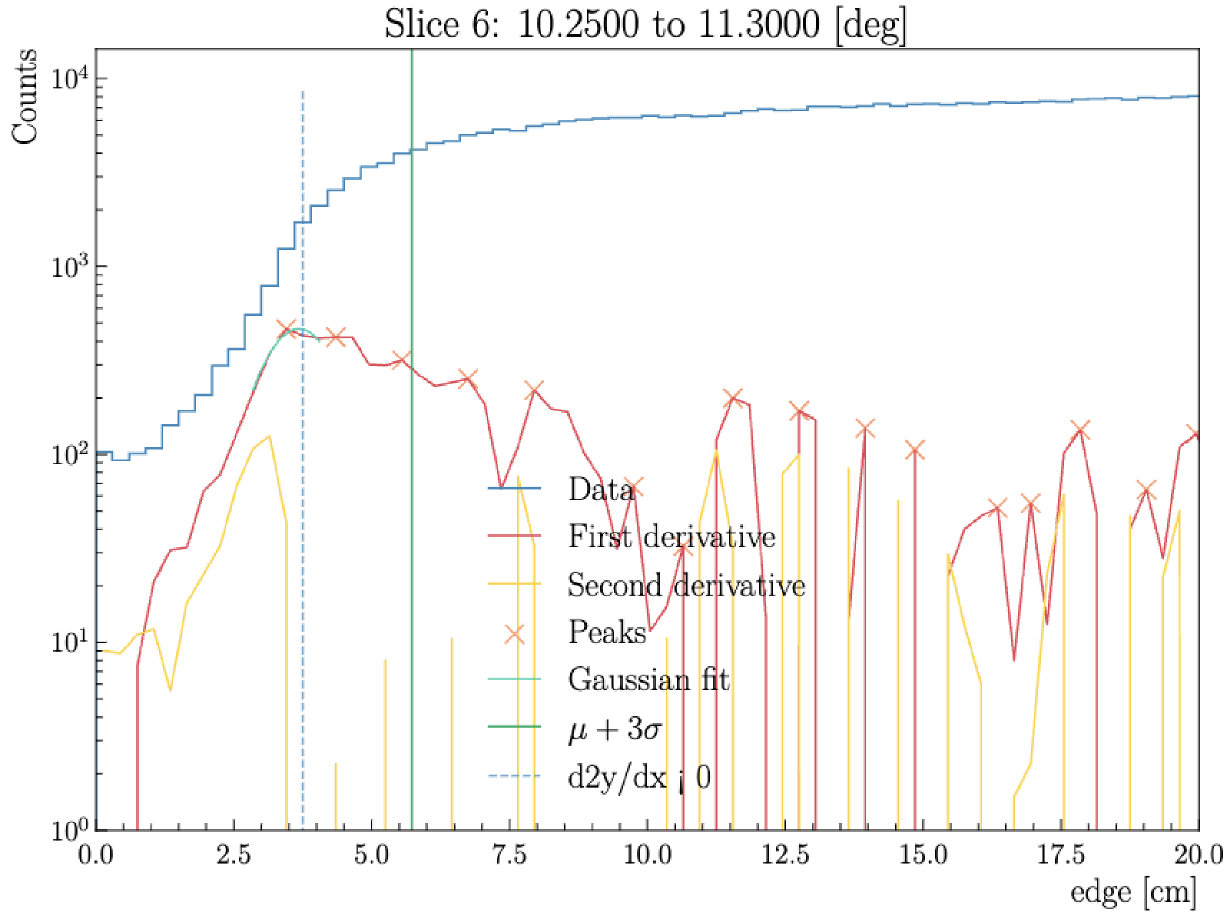
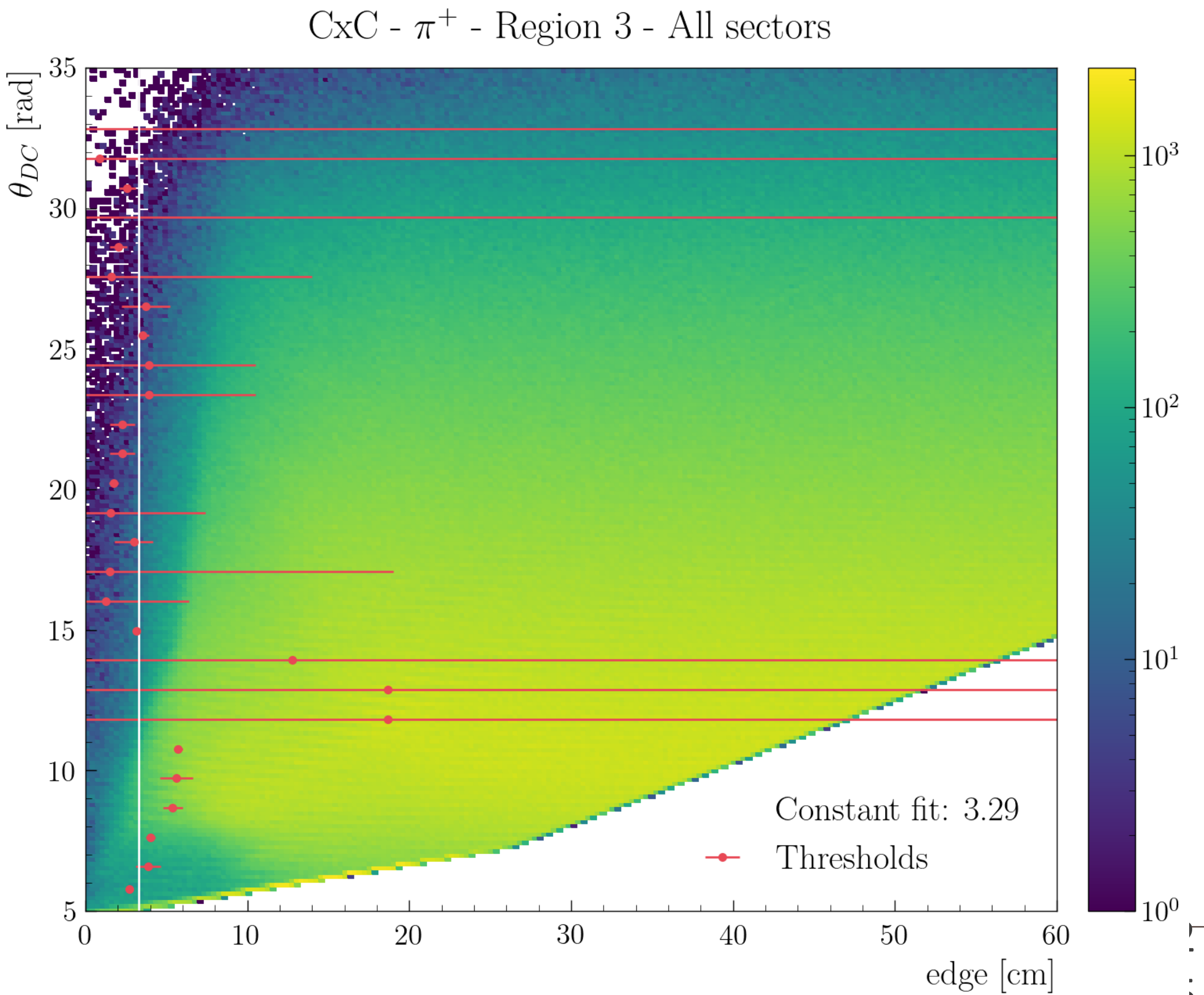
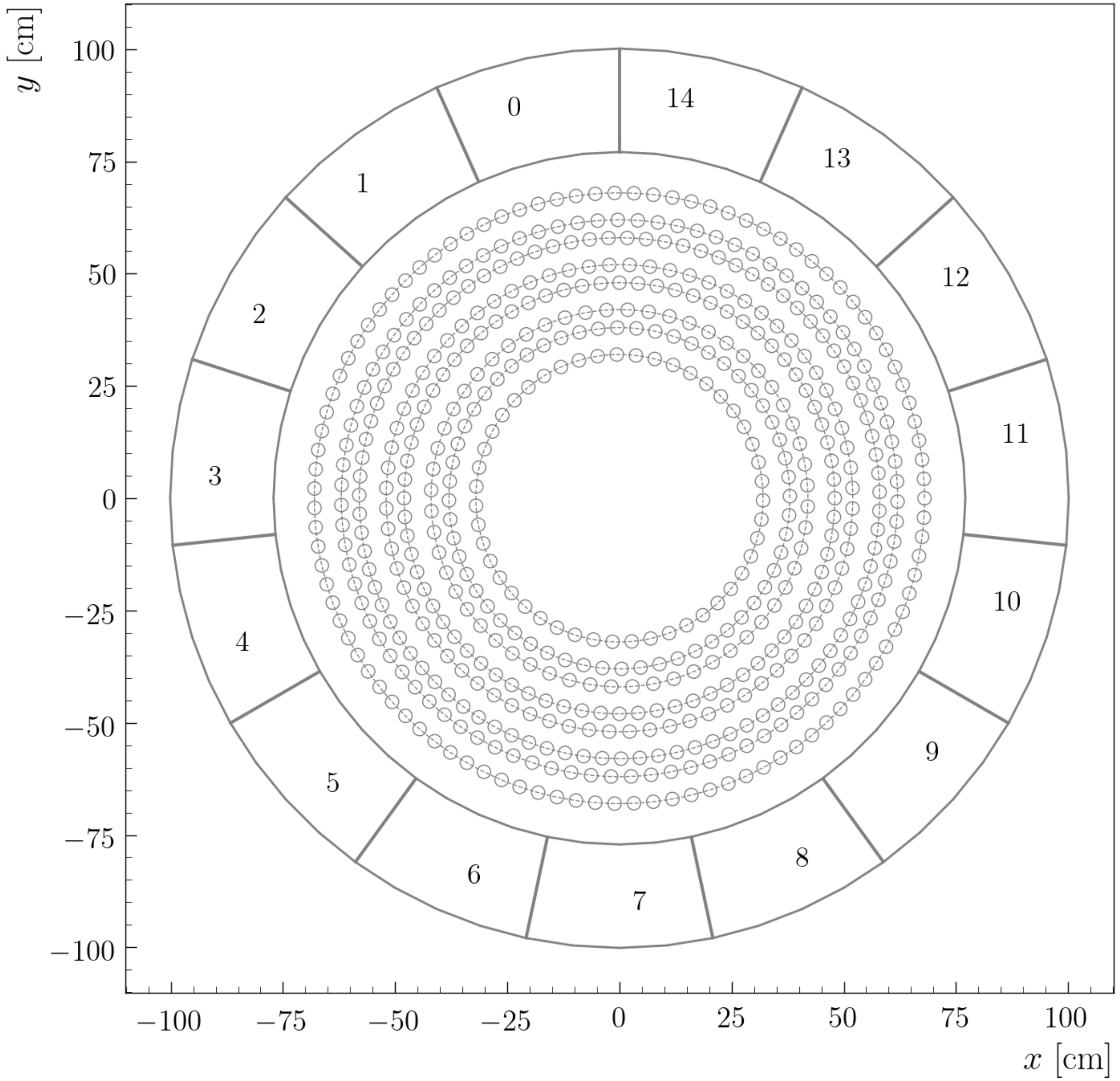




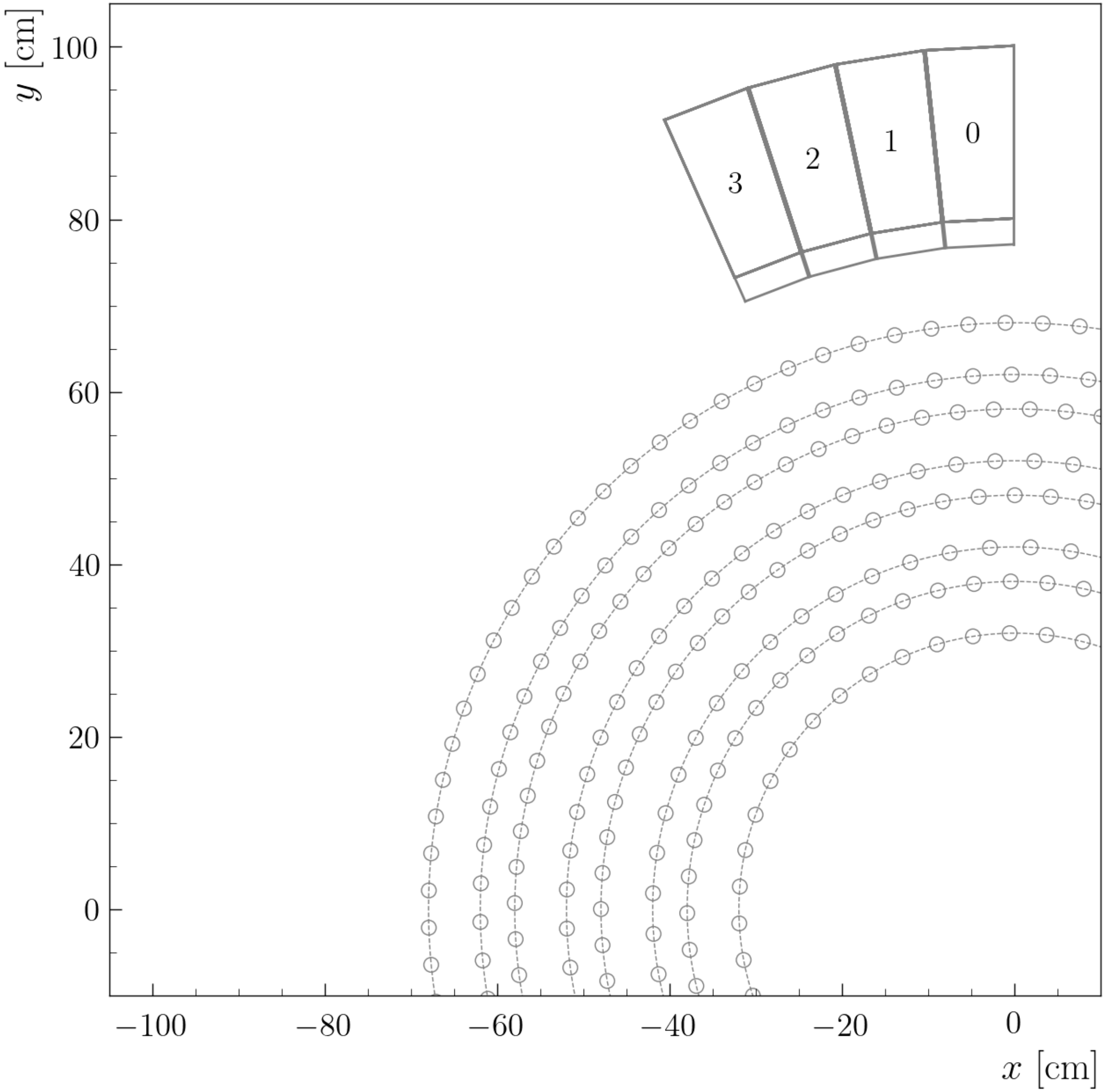
ATOF geometry

- The ATOF is composed of 15 phi-range/sectors numbered from 0 to 14.



ATOF geometry

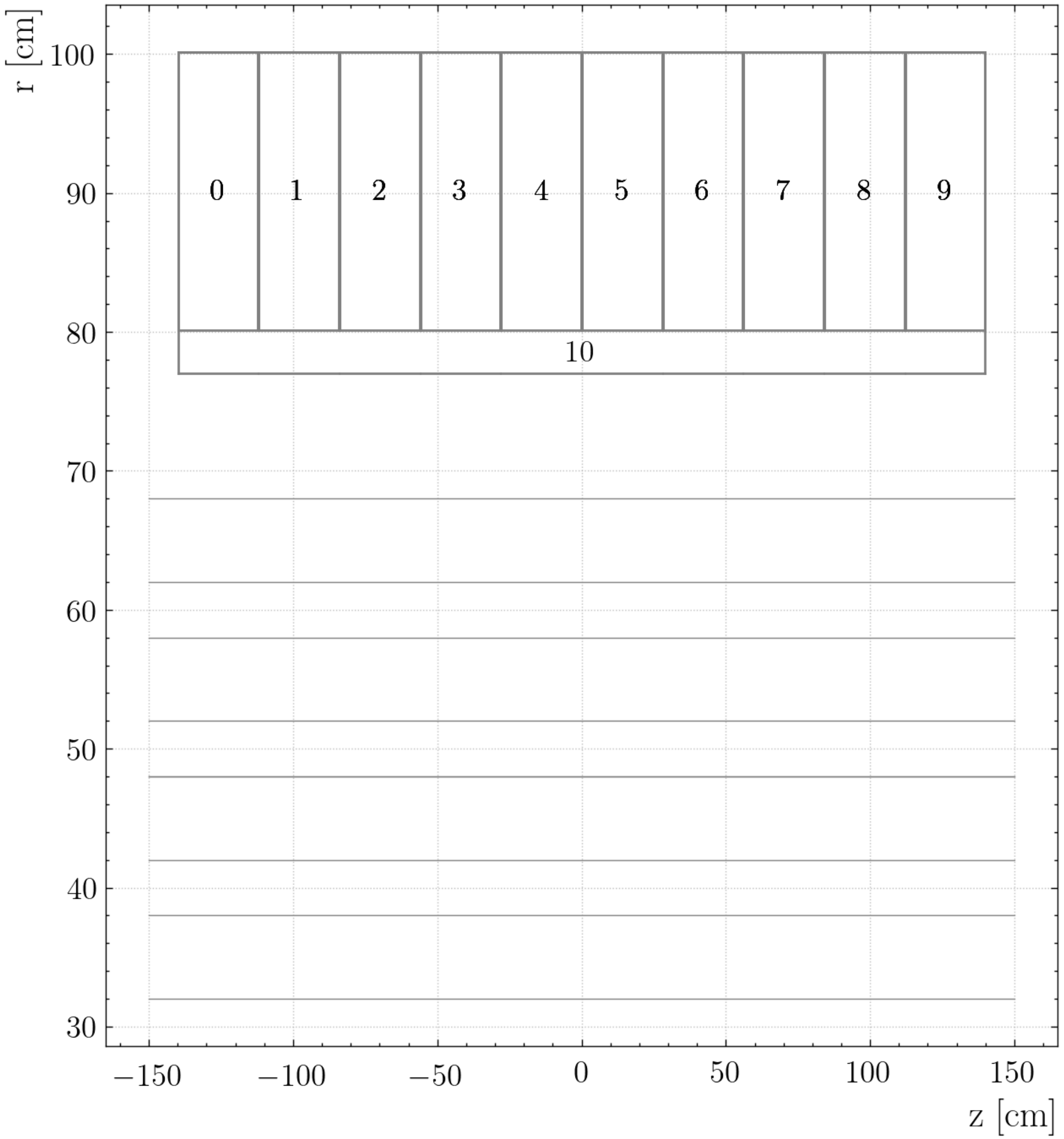
- The ATOF is composed of 15 phi-range/sectors numbered from 0 to 14.
- Each phi-range/sector is composed of 4 layers numbered from 0 to 3.



ATOF geometry

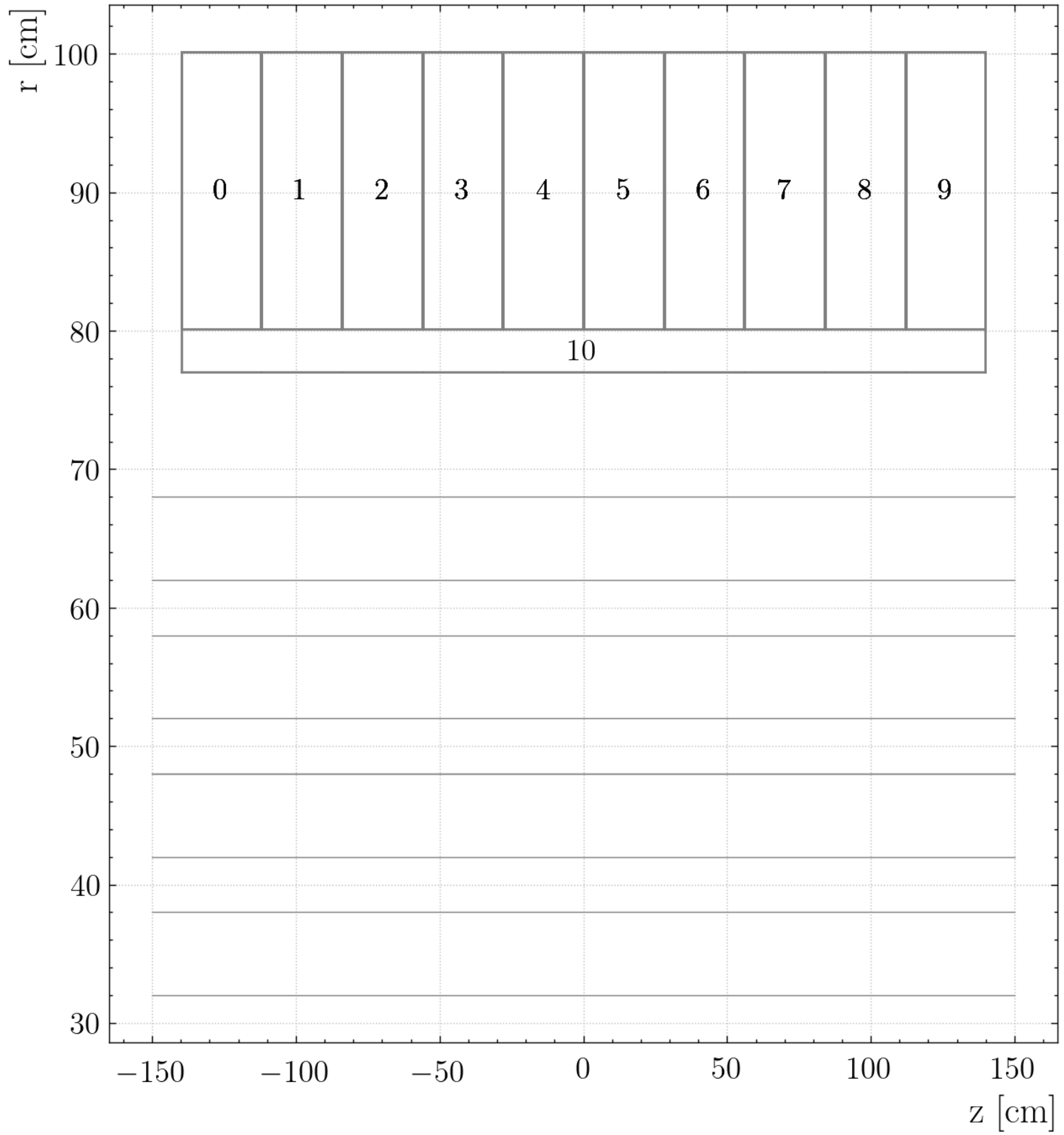
- The ATOF is composed of 15 phi-range/sectors numbered from 0 to 14.
- Each phi-range/sector is composed of 4 layers numbered from 0 to 3.
- Each layer is composed of 10 wedges numbered from 0 to 9 along the z-axis, and one bar numbered 10.
- In total, for each layer, there are 11 components.

- In total for the ATOF, there are 15*4*10 = 600 wedges and 15*4 = 60 bars.



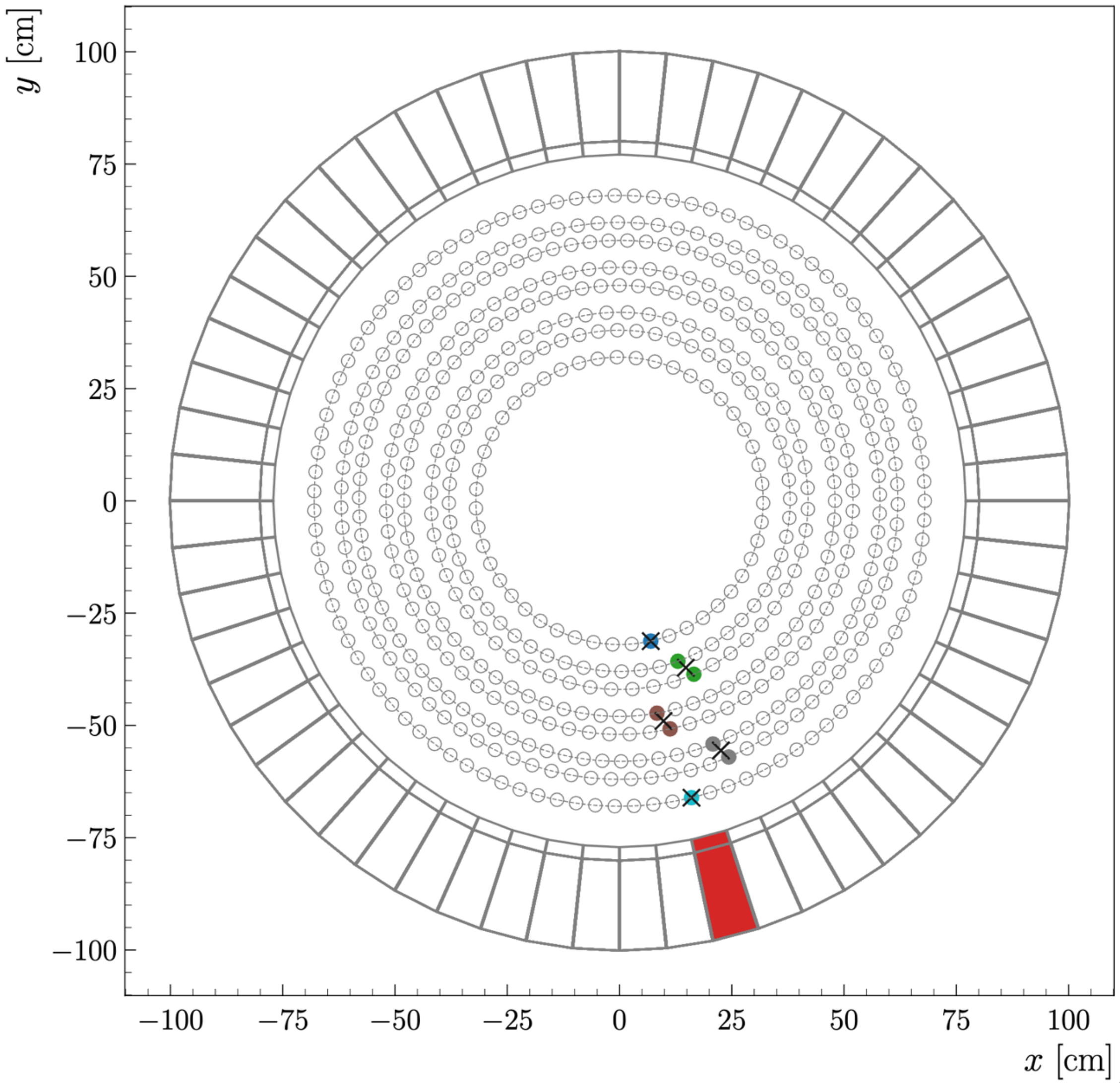
ATOF geometry

- So, to identify a specific wedge, I need a phi-range/sector, a layer , and a number between 0 and 9, inclusive.
- To identify a specific bar, I just need a phi-range/sector and a layer



AI for track matching and momentum prediction

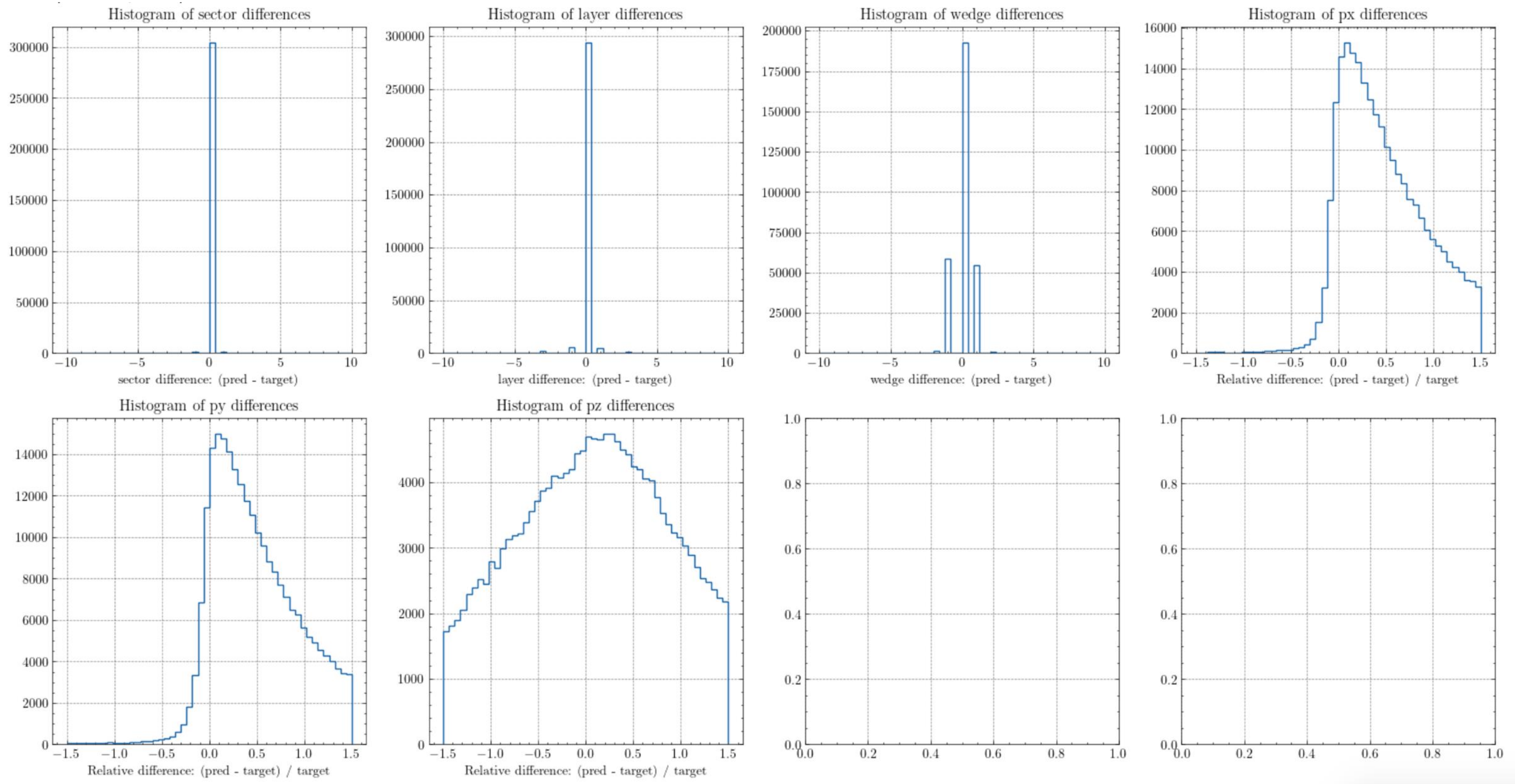
- Goal: predict with an AI which bar and edge should be hit based on the inter-clusters (or cluster)
- Inputs: 5 of 2D points (x, y)
(in total 10 numbers)
- Output: sector, layer, wedge, p_x, p_y, p_z



- For the training, I have generated 5M events and I select them with a good track (pred > 0.2) and only one bar and wedge

AI for track matching and momentum prediction

- I use an MLP with 10 inputs, 5 hidden layers of 512 neurons, and 6 outputs



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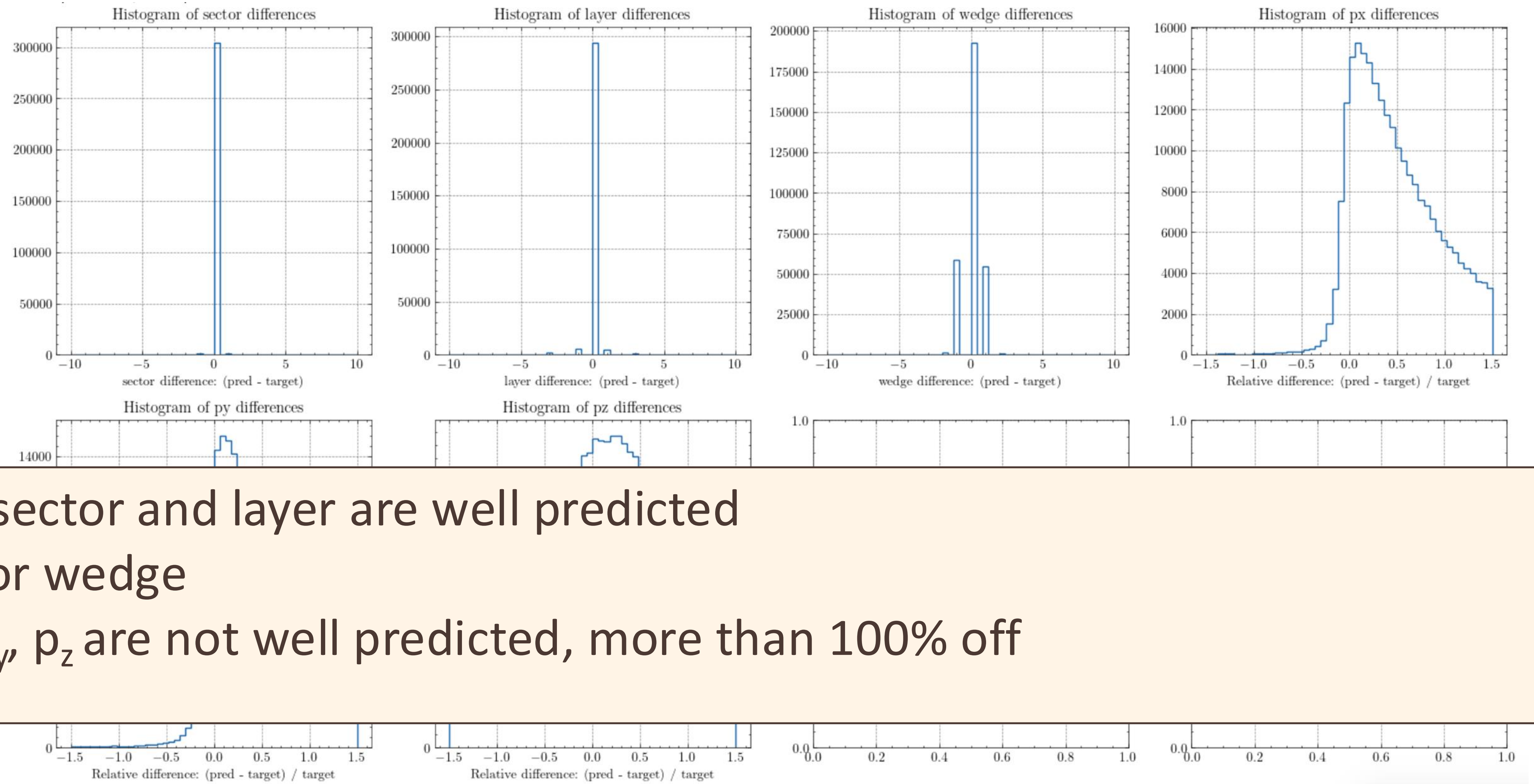


The sector and layer are well predicted

Ok for wedge

p_x, p_y, p_z are not well predicted, more than 100% off

AI for track matching and momentum prediction

- I have tried different architectures (number of neurons per layer, number of layers), increased the size of the training sample, created 10 networks of different sizes and averaged them, different loss functions (l1, mse, huber), …
- But I always end up with similar results.

- I have also tried to use the clusters (3D points) created by using the information of two layers with a different stereo-angle

- Loss function:
  - y_{pred} is the prediction of the model
  - y_{target} is the true value (from MC)

$$\mathrm{L1} = \left|y_{\mathrm{pred}} - y_{\mathrm{target}}\right|$$

$$\mathrm{MSE} = \left(y_{\mathrm{pred}} - y_{\mathrm{target}}\right)^2$$

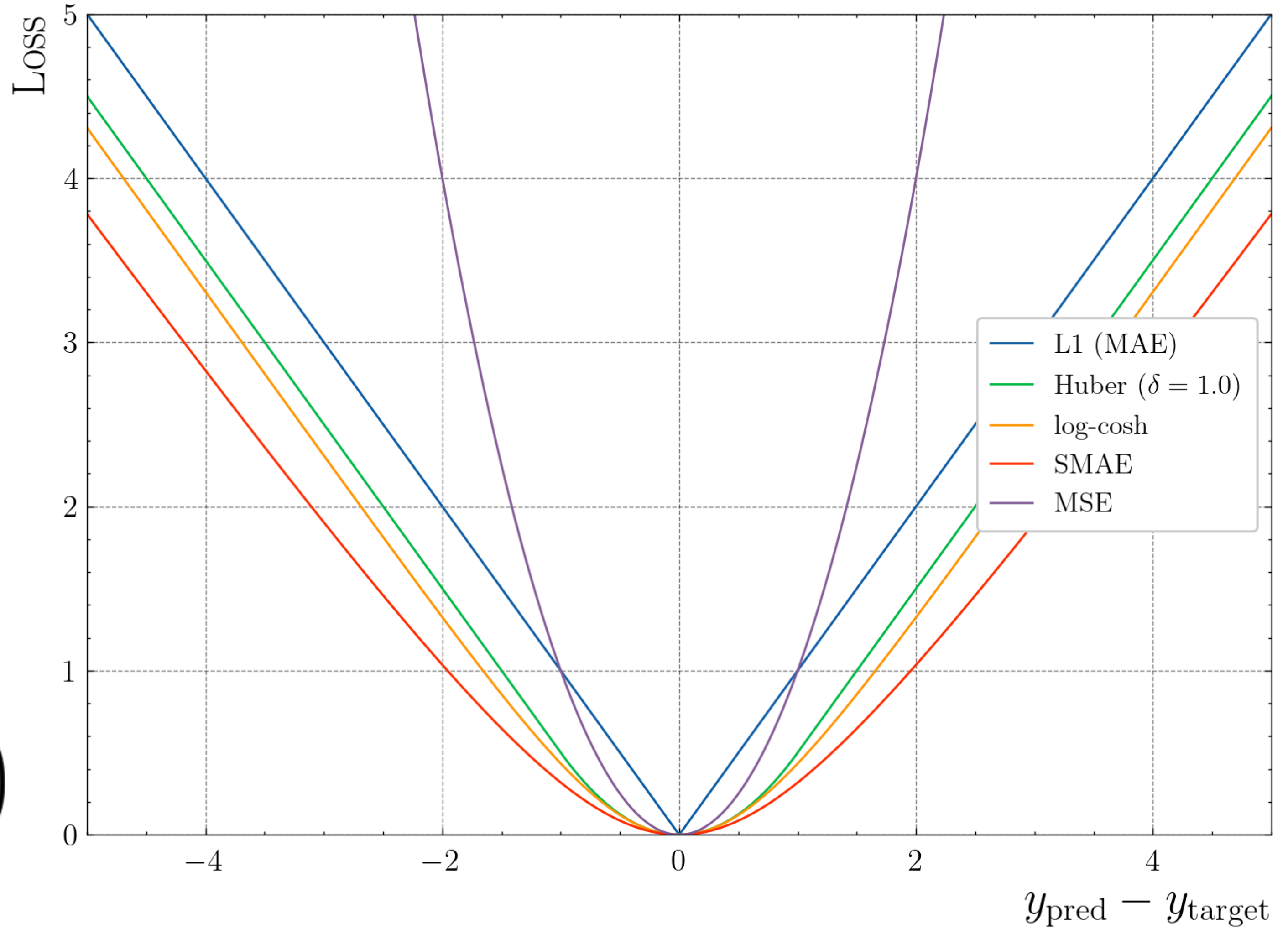
$$\mathrm{Huber} = \begin{cases} 0.5\left(y_{\mathrm{pred}} - y_{\mathrm{target}}\right)^2, & \text{if } \left|y_{\mathrm{pred}} - y_{\mathrm{target}}\right| < \delta \\ \delta \cdot \left(\left|y_{\mathrm{pred}} - y_{\mathrm{target}}\right| - 0.5 \cdot \delta\right), & \text{otherwise} \end{cases}$$

δ is a parameter

$$\text{log-cosh} = \log\left(\cosh\left(y_{\mathrm{pred}} - y_{\mathrm{target}}\right)\right)$$

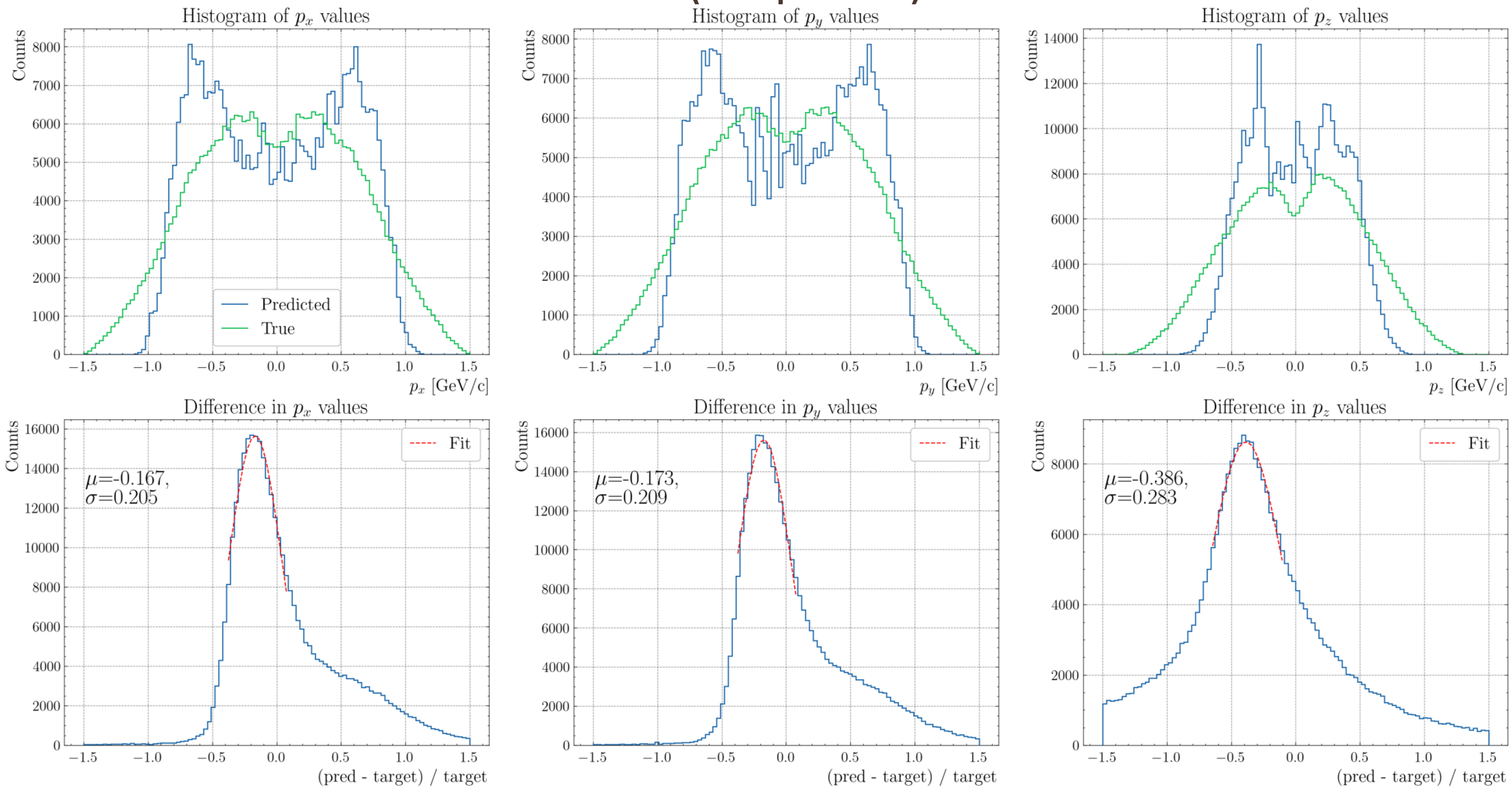
$$\mathrm{SMAE} = \left(y_{\mathrm{pred}} - y_{\mathrm{target}}\right)\tanh\left(\frac{y_{\mathrm{pred}} - y_{\mathrm{target}}}{2}\right)$$

Smooth MAE



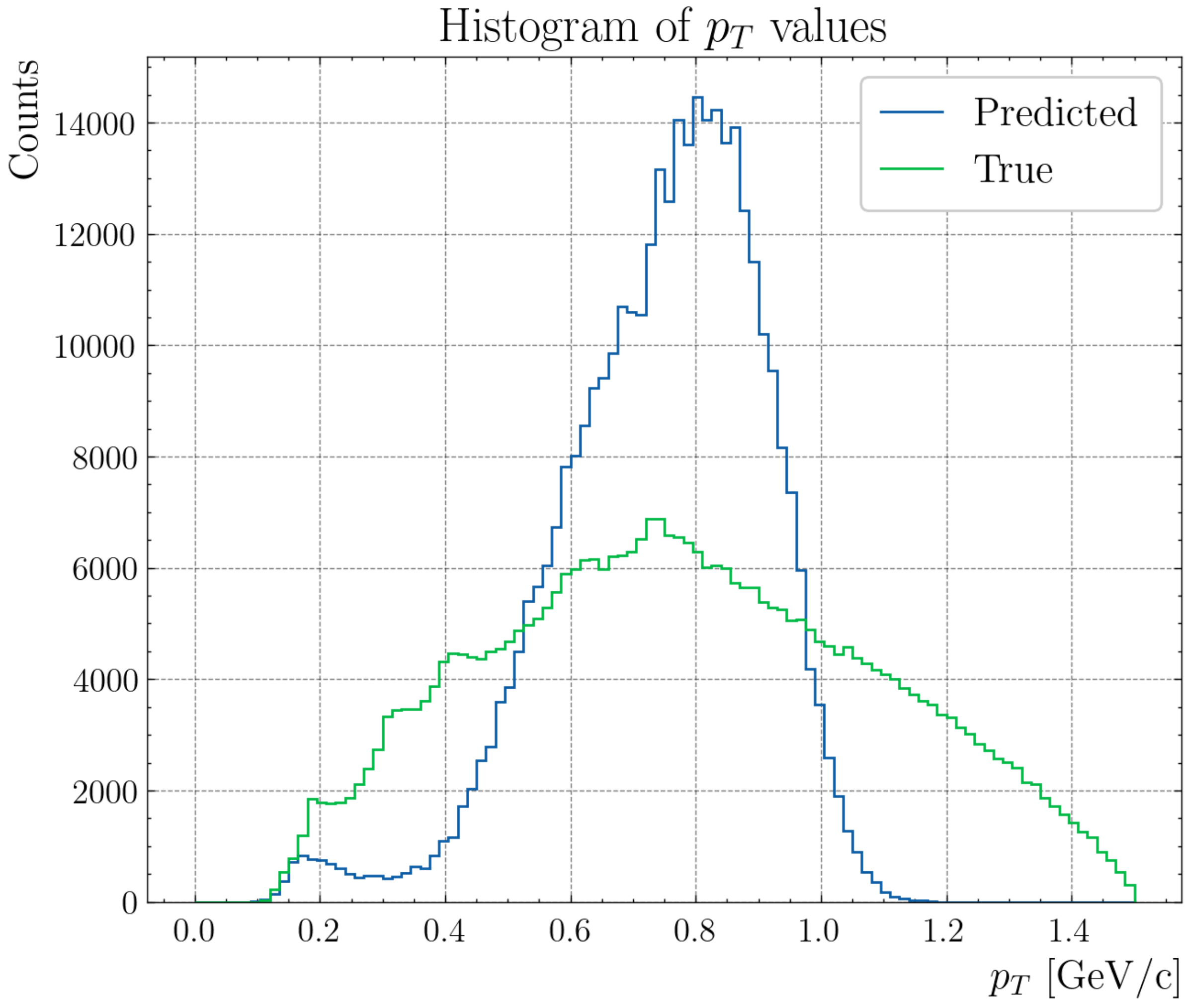
AI for track matching and momentum prediction

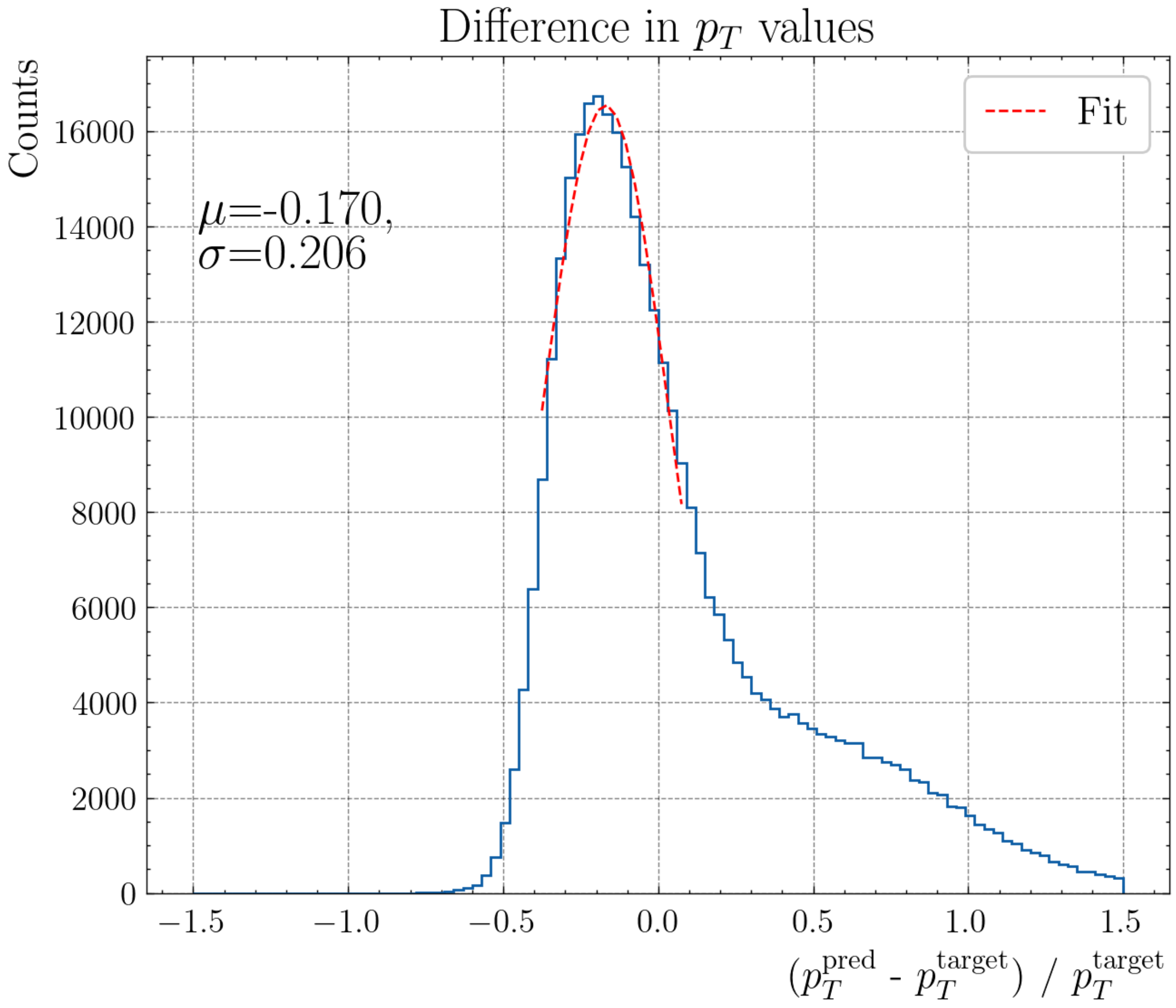
- I have also tried to use the cluster (3D points)



AI for track matching and momentum prediction

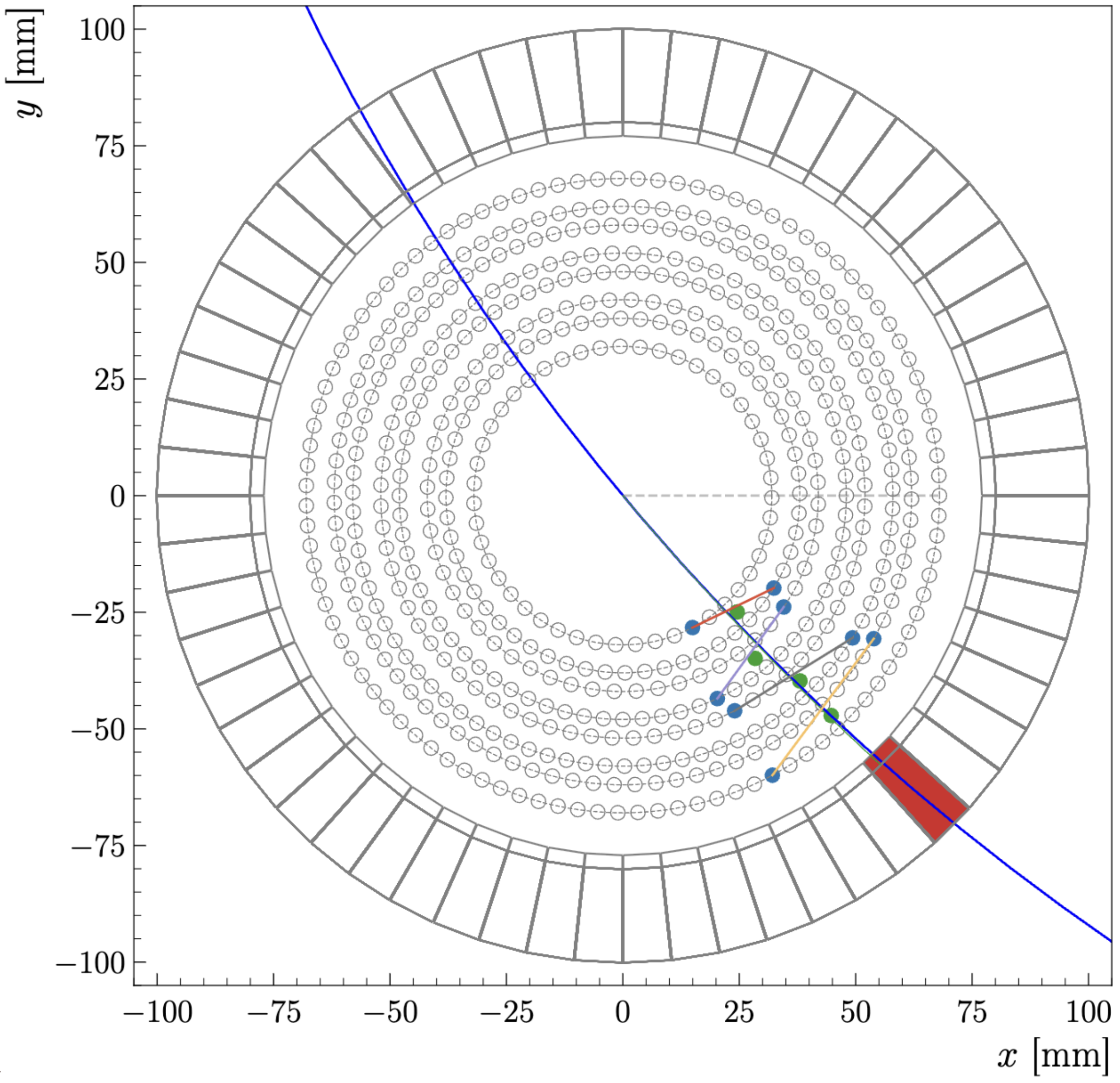
- Look to p_T for easier comparison with the conventional method

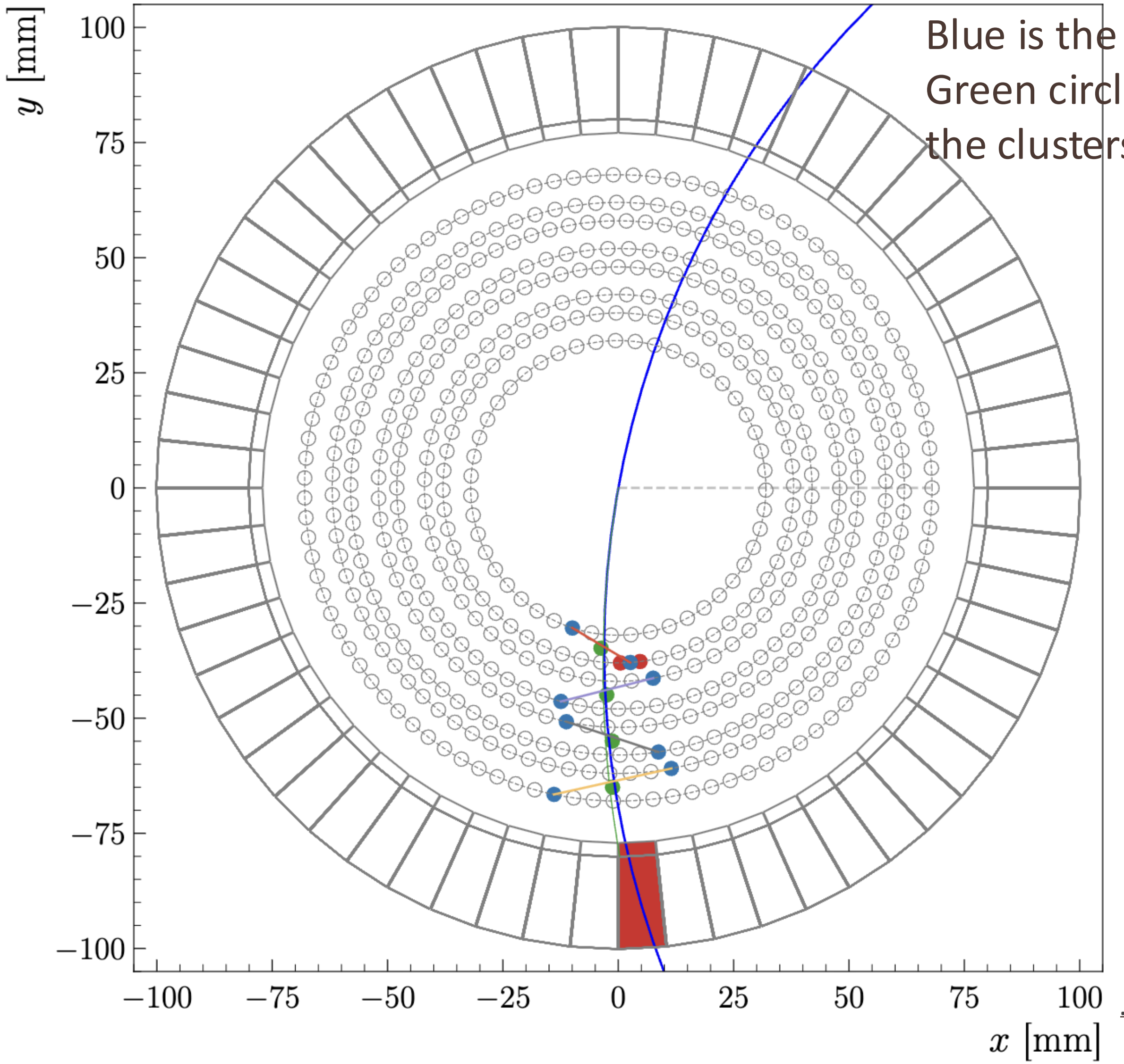




AI for track matching and momentum prediction

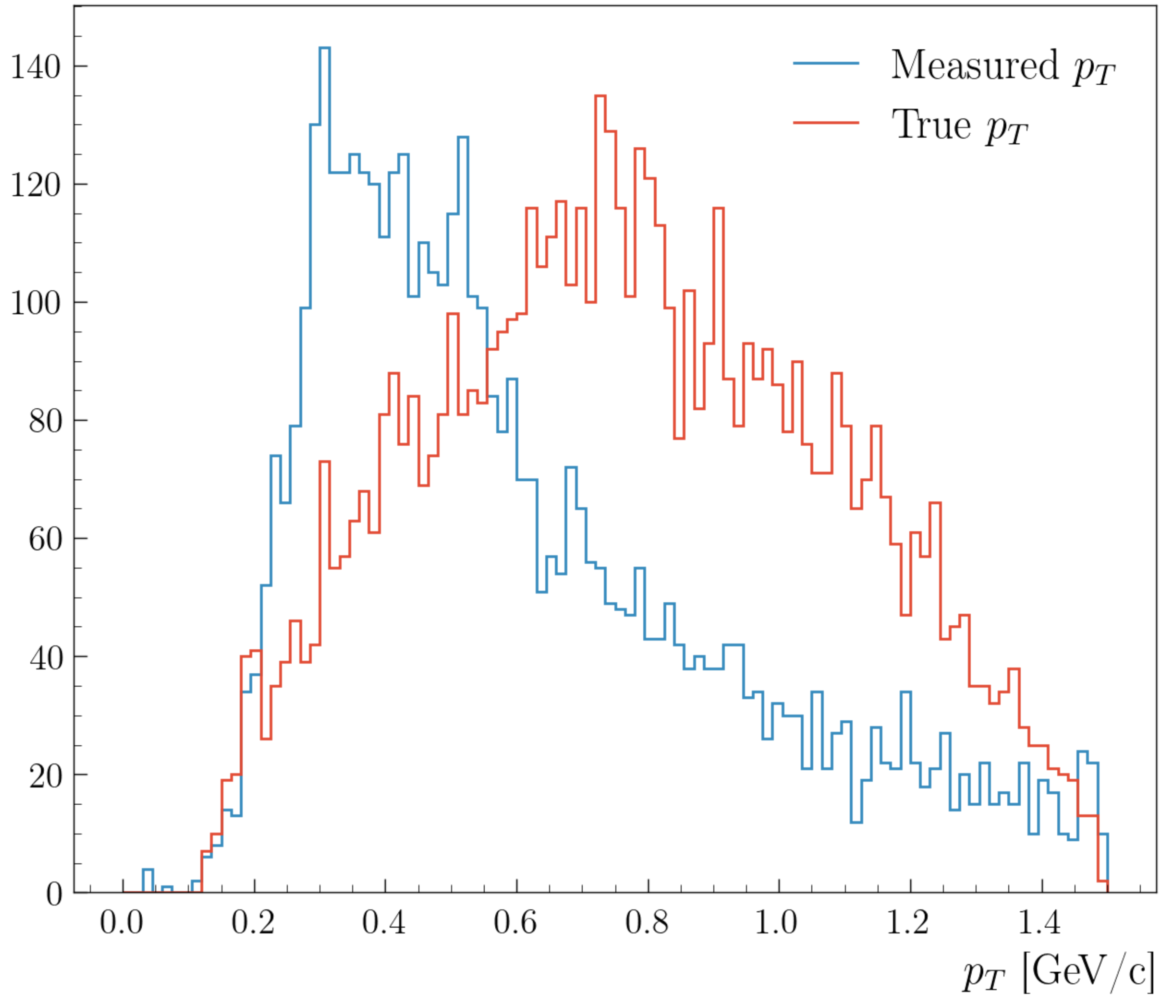
- By fitting the cluster (3D points) with a circle, we get the transverse momentum

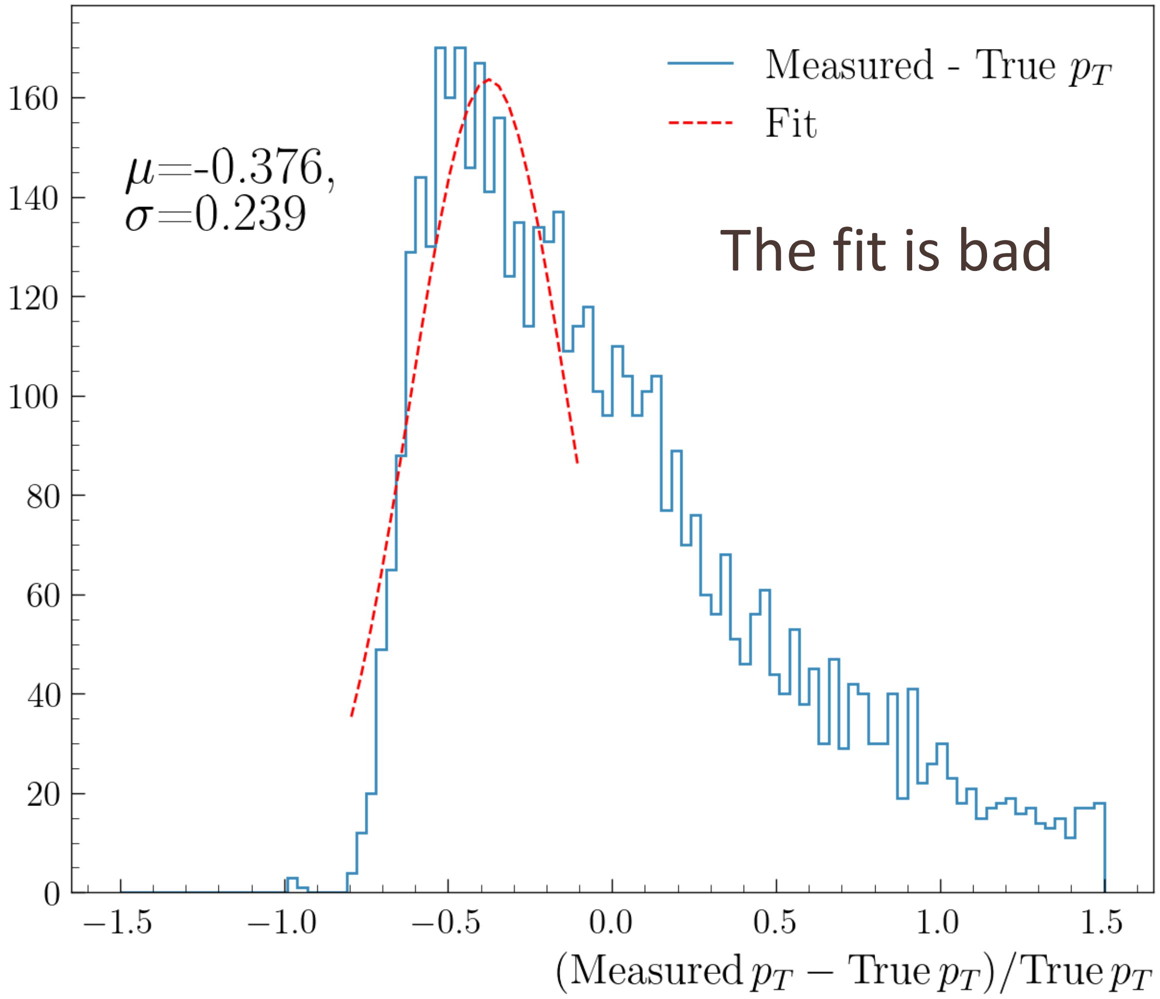




AI for track matching and momentum prediction

- By fitting the cluster (3D points) with a circle, we get the transverse momentum (low stats)





The fit is bad